NOTES ON STRAIGHTENING-UNSTRAIGHTENING AND THE ∞-CATEGORY OF ∞-CATEGORIES

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Abstract. A goal of mine for a while has been to try make sense of all the different types of fibrations laid out in Chapter 2 of [Lur09] and the various model structures they lead to and to understand the straightening and unstraightening constructions. These topics are very technical and maybe not so important to understand "in real life," but they are certainly necessary for really understanding Cat_∞ which I'd like to be less afraid of. Since I am at home for the holidays with a lot of extra time on my hands, I decided now is a good time to try and write up some notes on the subject. Here is the result of that effort.

Contents

1. Recollections on simplicial stuff 2
1.1. ∞-Categories as simplicial sets 2
1.2. Simplicial categories and topological categories 3
1.3. The simplicial nerve and $\mathfrak{C}[-]$ 4
1.4. Marked simplicial sets 5
2. Fibrations of simplicial sets 6
2.1. Lifting properties and elementary types of fibrations in Set_Δ 6
2.2. More on inner fibrations and correspondences of ∞-categories 7
2.3. Cartesian and coCartesian fibrations 10
2.4. Digression: mapping spaces 12
3. Model structures 13
3.1. Recollections on model categories 14
3.2. Some general constructions of model structures 15
3.3. Model structures on Set_Δ and Cat_Δ 16
3.4. Model structures on $(\mathsf{Set}_\Delta)_{/S}$ and a big comparison theorem 17
4. Straightening and unstraightening 18
4.1. The classical Grothendieck construction 18
4.2. Left fibrations and the Grothendieck construction 21
4.3. Straightening and unstraightening (unmarked) 21
4.4. Cat_∞ and the need for marked simplicial sets 24
4.5. Straightening and unstraightening (marked) 24
5. Applications to Cat_∞ 25
5.1. Classifying functors to Cat_∞ 25
5.2. Explicit construction of colimits (and limits) in Cat_∞ 25
6. Colimits 26
6.1. ∞-Categorical colimits as homotopy colimits 26
6.2. Relative colimits and left Kan extensions 26
Index 27
References 27

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1. Recollections on simplicial stuff

To begin, we recall various definitions from the theory of simplicial sets which we will make use of throughout these notes.

1.1. ∞-Categories as simplicial sets

By $[n]$ we denote the poset of the first $n+1$ natural numbers, $\{0, 1, \ldots, n\}$. By Δ we denote the category which has objects $[n]$ for each n and morphisms which are (non-strictly) order-preserving maps. A *simplicial object* in a category $\mathcal{C}$ is a functor $\Delta^{\mathsf{op}} \to \mathcal{C}$. dually, a *cosimplicial object* is a functor $\Delta \to \mathcal{C}$. We denote the functor category of simplicial objects in $\mathcal{C}$ by

$$\mathcal{C}_\Delta = \mathsf{Fun}(\Delta^{\mathsf{op}}, \mathcal{C}).$$

Dually, the functor category of cosimplicial objects in $\mathcal{C}$ is

$$\mathcal{C}^\Delta = \mathsf{Fun}(\Delta, \mathcal{C}).$$

Definition 1.1.1. A *simplicial set* is a simplicial object in Set, i.e., an object of Set_Δ.

Remark 1.1.2. It will sometimes be interesting to note that Set_Δ is actually the category of presheaves of sets over the category Δ. An immediate consequence is that Set_Δ is cocomplete, and we obtain a fully faithful embedding

$$y : \Delta \to \mathsf{Set}_\Delta,$$

the essential image of which generates Set_Δ under colimits.

Given a simplicial set S, we will write S_n for the set $S([n])$. The morphisms of Δ induce the following maps for each $n \geq 0$ and $j = 0, \ldots, n$:

(i) *face maps* $d_j : S_n \to S_{n-1}$, induced by the strictly order preserving map $[n-1] \to [n]$ which does not have j in its image, and
(ii) *degeneracy maps* $s_j : S_n \to S_{n+1}$, induced by the map $[n+1] \to [n]$ which is only not strictly order preserving at index j, over which the fiber is j and $j+1$.

Every order preserving map $[n] \to [m]$ factors as a composition of face and degeneracy maps, and so a simplicial set S is totally determined by the sets S_n for each $n \geq 0$, and the face and degeneracy maps.

Following [Lur09, A.2.7.2], for any finite linearly-ordered set J, we define the simplicial set Δ^J to be the representable functor

$$\Delta^J : [n] \mapsto \mathrm{Hom}_\Delta([n], J).$$

When $J = [n]$, we will write Δ^n in place of $\Delta^{[n]}$. Note that Δ^n may be thought of as the full subcategory of Δ comprising the objects $[0], \ldots, [n]$.

Remark 1.1.3. By the Yoneda lemma, for a simplicial set S, there is a natural identification

$$S_n \cong \mathrm{Hom}_{\mathsf{Set}_\Delta}(\Delta^n, S)$$

For any $0 \leq k \leq n$, define Λ^n_k to be the sub-simplicial set of Δ^n with

$$(\Lambda^n_k)_i = \{p : [i] \to [n] \text{ order preserving} : \{k\} \cup p([i]) \neq [n]\}.$$

That is, Λ^n_k coincides with Δ^n up to degree $n-2$; in degree $n-1$, the unique order-preserving map $p : [n-1] \to ([n] \setminus k)$ is omitted, and in degree n, $(\Lambda^n_k)_n$ is empty.

The simplicial set Λ^n_k is called the *kth horn of* Δ^n. When $k = 0, n$, Λ^n_k is called an *outer horn* of Δ^n, and if $0 < k < n$, Λ^n_k is called an *inner horn*.

This leads us to the following definition.

Definition 1.1.4 ([Lur09, 1.1.2.1, 1.1.2.4])**.** An ∞-*category* is a simplicial set $\mathcal{C}$ such that for each $0 < k < n$ and map of simplicial sets $f_0 : \Lambda^n_k \to \mathcal{C}$, there exists some f making the following diagram commute.

$$\begin{tikzcd} \Lambda^n_k \arrow[r, "f_0"] \arrow[d, hook] & \mathcal{C} \\ \Delta^n \arrow[ur, dashed, "f"'] & \end{tikzcd}$$

If, additionally, this property holds when k is allowed to be either 0 or n, then $\mathcal{C}$ is a *Kan complex*, or ∞*-groupoid*.

Interpreting the previous definition with regards to Remark 1.1.3, *objects* of an ∞-category $\mathcal{C}$ correspond to functors $(\Delta^0)^{\mathsf{op}} \to \mathcal{C}$, *morphisms* correspond to functors $(\Delta^1)^{\mathsf{op}} \to \mathcal{C}$, 2*-morphisms* to functors $(\Delta^2)^{\mathsf{op}} \to \mathcal{C}$, and so on.

Example 1.1.5. When $n = 2$ and $k = 1$, the "horn filling property" of Definition 1.1.4 becomes the condition that, for any three objects a, b, c in $\mathcal{C}$ and morphisms $f : a \to b$ and $g : b \to c$, there exists some "composition" h and "homotopy" α (i.e., an element of $\mathcal{C}_2$ with faces a, b, and c) filling in the following triangle

$$\begin{tikzcd} & b \arrow[dr, "g"] \arrow[d, phantom, "\Downarrow\alpha"] & \\ a \arrow[ur, "f"] \arrow[rr, dashed, "h"'] & {} & c \end{tikzcd}$$

For $\mathcal{C}$ to be an ∞-groupoid, we also ask for the following outer horn filling properties.

$$\begin{tikzcd} & b \arrow[dr, "g"] \arrow[d, phantom, "\Downarrow\alpha'"] & \\ a \arrow[ur, dashed, "f"] \arrow[rr, "h"'] & {} & c \end{tikzcd} \qquad \begin{tikzcd} & b \arrow[dr, dashed, "g"] \arrow[d, phantom, "\Downarrow\alpha''"] & \\ a \arrow[ur, "f"] \arrow[rr, "h"'] & {} & c \end{tikzcd}$$

Taking $c = a$ and h to be the identity, these conditions are equivalent to asking each g and f to have right and left inverses (up to a choice α' and α'' of homotopy), respectively.

The following result shows that for $\mathcal{C}$ to be a Kan complex, it's enough that it satisfies *either* the left or right horn filling properties.

Proposition 1.1.6 ([Lur09, 1.2.5.1])**.** *Let $\mathcal{C}$ be a simplicial set. The following are equivalent.*

(i) The simplicial set $\mathcal{C}$ is an ∞-groupoid and its homotopy category $\mathsf{h}\mathcal{C}$ is a groupoid;
(ii) the simplicial set $\mathcal{C}$ satisfies the extension property for all horn inclusions $\Lambda^n_i \hookrightarrow \Delta^n$ with $0 \le i < n$;
(iii) the simplicial set $\mathcal{C}$ satisfies the extension property for all horn inclusions $\Lambda^n_i \hookrightarrow \Delta^n$ with $0 < i \le n$;
(iv) the simplicial set $\mathcal{C}$ is a Kan complex.

Proposition 1.1.7. *A classical intuition for the previous proposition is the fact that any (ordinary) category in which every morphism admits a retract (or, dually, a section) is a groupoid.*

1.2. Simplicial categories and topological categories

Let us solidify some terminology. In the previous section, we defined ∞-categories to be simplicial sets satisfying the inner horn filling property. In this section, we treat the related topic of simplicial categories and topological categories.

Definition 1.2.1 ([Lur09, 1.1.4.1])**.** A *simplicial category* is a category which is enriched over Set_Δ.

Remark 1.2.2 ([Lur09, 1.1.4.2])**.** Every simplicial category corresponds to a simplicial object of Cat. Conversely, the simplicial objects of Cat which arise from a simplicial category are exactly those $C : \Delta^{\mathsf{op}} \to \mathsf{Cat}$ for which the underlying category of objects, $C([0])$ is constant as a simplicial set (i.e., $C([0])_n = *$) for all $n \ge 0$. Nevertheless, we will denote the category of simplicial categories by Cat_Δ.

Definition 1.2.3 ([Lur09, 1.1.1.6])**.** A *topological category* is a category which is enriched over the category Top of (compactly generated, weakly Hausdorff) topological spaces. We denote the category of topological categories by $\mathsf{Cat}_{\mathrm{Top}}$

Remark 1.2.4. The choice to consider compactly generated, weakly Hausdorff topological spaces above is to facilitate some equivalence theorems later. We won't need to worry about it again.

Definition 1.2.5 ([Lur09, 1.1.3.6])**.** A functor $F : \mathcal{C} \to \mathcal{D}$ of topological categories is an *equivalence of topological categories* if

(i) F is essentially surjective up to homotopy: for every $d \in \mathcal{D}$, there is some $c \in \mathcal{C}$ and homotopy equivalence $f : F(c) \to d$ in $\mathcal{D}$ (that is, f is invertible up to homotopy);

(ii) for every pair of objects $c, c' \in \mathcal{C}$, the induced map

$$F : \mathsf{Maps}_{\mathcal{C}}(c, c') \to \mathsf{Maps}_{\mathcal{D}}(F(c), F(c'))$$

is a weak homotopy equivalence of (topological) spaces.

It is often possible to go back and forth between the language of simplicial categories, topological categories, and ∞-categories, as we will see below.

Recall that there is an adjunction

$$|-| : \mathsf{Set}_\Delta \rightleftarrows \mathsf{Top} : \mathrm{Sing}$$

between the geometric realization and singular chains functors. By that one theorem of Milnor, both of these functors preserve finite limits, and so we can apply them to mapping objects to go back and forth between simplicial and topological categories. Indeed, the unit and counits induce maps $\mathcal{C} \to \mathrm{Sing}\,|\,\mathcal{C}\,|$ and $|\,\mathrm{Sing}\,\mathcal{D}\,| \to \mathcal{C}$ which are equivalences at the level of homotopy categories. See [Lur09, 1.1.4] for more discussion.

1.3. **The simplicial nerve and $\mathfrak{C}[-]$**

Recall the following classical definition.

Definition 1.3.1. The (ordinary) *nerve* of an ordinary category $\mathcal{C}$ is the simplicial set $N(\mathcal{C})$ characterized by the relation

$$\mathrm{Hom}_{\mathsf{Set}_\Delta}(\Delta^n, N(\mathcal{C})) = \mathrm{Hom}_{\mathsf{Cat}}([n], \mathcal{C}).$$

This construction is perfectly valid, but not so interesting if $\mathcal{C}$ is a simplicial category since it does not make use of the positive-dimensional cells of the mapping objects in $\mathcal{C}$. We will modify this definition by producing a "simplicial thickening" functor $\mathfrak{C}[-]$, to which the simplicial nerve functor (sometimes called the "homotopy-coherent nerve") will be right adjoint.

The first step is to define simplicial categories $\mathfrak{C}[\Delta^n]$ to take the place of $[n]$ in Definition 1.3.1. For an explicit general definition of $\mathfrak{C}[\Delta^J]$ for any linearly ordered set J, see [Lur09, 1.1.5.1, 1.1.5.2]. Since we will only have need of $\mathfrak{C}[\Delta^n]$ (and the explicit definition is somewhat unenlightening), we "define" it via example as follows.

Example 1.3.2. Let us consider $n = 3$, so that the poset category $[3]$ can be drawn as

$$\begin{tikzcd}
0 \arrow[r, "p_{01}"] \arrow[rr, bend right, "p_{02}"] \arrow[rrr, bend right=60, "p_{03}"'] & 1 \arrow[r, "p_{12}"] \arrow[rr, bend left, "p_{13}"] & 2 \arrow[r, "p_{23}"] & 3
\end{tikzcd}$$

The simplicial category $\mathfrak{C}[\Delta^3]$ then has the same objects as $[3]$, but the morphism simplicial sets in $\mathfrak{C}[\Delta^3]$ track all the different possible compositions in $[3]$.

For example, the simplicial set $\mathsf{Maps}_{\mathfrak{C}[\Delta^3]}(i, i)$ is a single 0-cell corresponding to id_i for each $i = 0, 1, 2, 3$; the simplicial set $\mathsf{Maps}_{\mathfrak{C}[\Delta^3]}(0, 2)$ is given by (the 1-simplex)

$$p_{12} \circ p_{01} \text{ ——— } p_{02},$$

$\mathsf{Maps}_{\mathfrak{C}[\Delta^3]}(0, 3)$ is given by (the "filled-in" 3-simplex)

$$\begin{tikzcd}
 & p_{13} \circ p_{01} \arrow[dl, dash] \arrow[dr, dash] \arrow[dd, dash] & \\
p_{23} \circ p_{12} \circ p_{01} \arrow[rr, dash] \arrow[dr, dash] & & p_{03} \arrow[dl, dash] \\
 & p_{23} \circ p_{02} &
\end{tikzcd}$$

and $\mathsf{Maps}_{\mathfrak{C}[\Delta^3]}(i, j)$ is empty whenever $i > j$.

To summarize, $\mathfrak{C}[\Delta^n]$ is a simplicial category with the same objects as $[n]$ and with homotopy equivalent mapping spaces (after geometric realization):

$$|\,\mathsf{Maps}_{\mathfrak{C}[\Delta^n]}(i,j)| \simeq |\,\mathsf{Maps}_{[n]}(i,j)|$$

for all $i, j \in [n]$. The point is that the vertices of $\mathsf{Maps}_{\mathfrak{C}[\Delta^n]}(i,j)$ correspond to the different equivalent compositions of morphisms from i to j in $[n]$, and the higher dimensional cells witness their homotopy equivalence, up to all higher coherences. That is, $\mathfrak{C}[\Delta^n]$ is exactly $[n]$ where the property of strict associativity has been weakened to associativity up to coherent homotopy.

Thus, we have defined a functor

$$\begin{aligned} \Delta &\to \mathsf{Cat}_\Delta \\ \Delta^n &\mapsto \mathfrak{C}[\Delta^n]. \end{aligned} \tag{1.3.3}$$

To define $\mathfrak{C}[-]$ on the entire category Set_Δ, we leverage the following general construction of left Kan extensions along the Yoneda embedding $y : \Delta \to \mathsf{Set}_\Delta$ detailed in [Rhi14, 1.5.1].

Construction 1.3.4. Define

$$\mathfrak{C}[-] : \mathsf{Set}_\Delta \to \mathsf{Cat}_\Delta$$

be the left Kan extension of the functor $\Delta \to \mathsf{Cat}_\Delta$ defined in (1.3.3) along the Yoneda embedding:

$$\begin{tikzcd} \Delta \arrow[r] \arrow[d, hook, "y"'] & \mathsf{Cat}_\Delta \\ \mathsf{Set}_\Delta \arrow[ur, dashed, "\mathfrak{C}[-]"'] & \end{tikzcd}$$

By construction, $\mathfrak{C}[-]$ is colimit preserving and so admits a right adjoint,

$$N : \mathsf{Cat}_\Delta \to \mathsf{Set}_\Delta$$

which we define to be the *simplicial nerve* (sometimes, *homotopy-coherent nerve*) functor. As a consequence of this construction, we obtain the following formula for N. Given a simplicial category $\mathcal{C}$, we have

$$N(\mathcal{C})_n = \mathrm{Hom}_{\mathsf{Set}_\Delta}(\Delta^n, N(\mathcal{C})) = \mathrm{Hom}_{\mathsf{Cat}_\Delta}(\mathfrak{C}[\Delta^n], \mathcal{C}).$$

Indeed, this can be taken as the definition of N, as in [Lur09, 1.1.5.5].

Remark 1.3.5. Compare Definition 1.3.1 of the classical nerve to the definition above of the simplicial nerve. The latter is obtained from the former by replacing $[n]$ with $\mathfrak{C}[\Delta^n]$, and taking Hom in Cat_Δ, rather then in Cat. These changes capture the simplicial structure of $\mathcal{C}$ which the ordinary nerve forgets.

1.4. **Marked simplicial sets**

For later, we will record some basic definitions about the category of marked simplicial sets, Set_Δ^+.

Definition 1.4.1 ([Lur09, 3.1.0.1])**.** A *marked simplicial set* is a pair $(X, \mathcal{E})$ where X is a simplicial set and $\mathcal{E} \subset X_1$ is a subset of *marked* edges in X which contains every degenerate edge.

A morphism of marked simplicial sets $f : (X, \mathcal{E}) \to (X', \mathcal{E}')$ is a map of simplicial sets $f : X \to X'$ such that $f(\mathcal{E}) \subset \mathcal{E}'$. We denote the category of marked simplicial sets by Set_Δ^+.

Example 1.4.2. For every simplicial set S, there are several special examples of marked simplicial sets which we will be concerned with:

(i) $S^\flat$ in which the marked edges comprise the set $s_0(S_0)$ of degenerate edges only;
(ii) $S^\sharp$ in which the marked edges comprise the set S_1 of all edges of S.

Remark 1.4.3. For brevity, we will write $(\mathsf{Set}_\Delta^+)_{/S}$ to mean $(\mathsf{Set}_\Delta^+)_{/S^\sharp}$ for a given simplicial set S.

Example 1.4.4 ([Lur09, 3.1.1.9])**.** If $p : X \to S$ is a coCartesian fibration (Definition 2.3.9), then by $X^\natural$ be denote the marked simplicial set $(X, \mathcal{E})$ where $\mathcal{E}$ is the set of p-coCartesian edges in X.

Remark 1.4.5 (Cartesian closed structure [Lur09, 3.1.3])**.** The category of marked simplicial sets is Cartesian closed. That is, for every $X, Y \in \mathsf{Set}_\Delta^+$, there is a mapping object $Y^X \in \mathsf{Set}_\Delta^+$ such that the functor $(-)^X : \mathsf{Set}_\Delta^+ \to \mathsf{Set}_\Delta^+$ is right adjoint to $(-) \times X$. The counit of this adjunction provides a natural "evaluation map:"

$$Y^X \times X \to Y.$$

Define $\mathsf{Maps}^\flat(X, Y)$ to be the underlying simplicial set of the marked simplicial set Y^X, and let $\mathsf{Maps}^\sharp(X, Y) \subset \mathsf{Maps}^\flat(X, Y)$ be the simplicial subset containing those simplices $\sigma \in Y^X$ with the property that every edge of σ is marked.

When working in $(\mathsf{Set}_\Delta^+)_{/S}$, one can make sense of subsets $\mathsf{Maps}_S^\flat(X, Y) \subset \mathsf{Maps}^\flat(X, Y)$ and $\mathsf{Maps}_S^\sharp(X, Y) \subset \mathsf{Maps}^\sharp(X, Y)$ as the subsets of maps over S.

Proposition 1.4.6 ([Lur09, 3.1.3.1])**.** *Given any $X \in (\mathsf{Set}_\Delta^+)_{/S}$ and $Y \to S$ a Cartesian fibration (Definition 2.3.4), $\mathsf{Maps}_S^\flat(X, Y^\natural)$ is an ∞-category, and $\mathsf{Maps}_S^\sharp(X, Y^\natural)$ is the maximal subset which is a Kan complex.*

2. Fibrations of simplicial sets

A lot of the difficulties of understanding the underlying simplicial theory of ∞-categories come from keeping the various types of fibrations of simplicial sets and their corresponding lifting properties straight. Indeed, all of Chapter 2 of [Lur09] is dedicated to this. As an outline, we follow this very handy implication diagram from [Lur09, 2.0.0.5].

$$\begin{tikzcd}
 & \text{Trivial Fibration} \arrow[d, Rightarrow] & \\
 & \text{Kan Fibration} \arrow[dl, Rightarrow] \arrow[dr, Rightarrow] & \\
\text{Left Fibration} \arrow[d, Rightarrow] & & \text{Right Fibration} \arrow[d, Rightarrow] \\
\text{coCartesian Fibration} \arrow[dr, Rightarrow] & & \text{Cartesian Fibration} \arrow[dl, Rightarrow] \\
 & \text{Categorical Fibration} \arrow[d, Rightarrow] & \\
 & \text{Inner Fibration} &
\end{tikzcd}$$

2.1. Lifting properties and elementary types of fibrations in Set_Δ

Our various notions of fibrations all have to do with the right lifting property, which we define below. Dually, we will sometimes care about cofibrations, which concern the left lifting property (although in some cases, a more elementary definition is possible).

Definition 2.1.1 ([Lur09, A.1.2])**.** Let $p : a \to b$ and $q : c \to d$ be morphisms in a category $\mathcal{C}$. We say p has the *left lifting property with respect to* q and, dually, q has the *right lifting property with respect to* p if, given the following commuting square involving p and q there exists some *lift* $\ell : b \to c$ making the following commute:

$$\begin{tikzcd}
a \arrow[r] \arrow[d, "p"'] & c \arrow[d, "q"] \\
b \arrow[r] \arrow[ur, dashed, "\ell"] & d.
\end{tikzcd}$$

We make our first definition of a type of fibration as follows.

Definition 2.1.2 ([Lur09, 2.0.0.2])**.** Let $f : X \to S$ be a morphism of simplicial sets. Then if f satisfies the right lifting property

$$\begin{tikzcd}
\partial\Delta^n \arrow[r, "g"] \arrow[d, hook] & X \arrow[d, "f"] \\
\Delta^n \arrow[r, "\bar{g}"'] \arrow[ur, dashed] & S
\end{tikzcd}$$

for every n and maps g and $\bar{g}$, then f is called a *trivial Kan fibration* (or sometimes just *trivial fibration*).

Trivial Kan fibrations will take the place of "homotopy equivalences" in the following sense.

Proposition 2.1.3 ([Lur25, 3.3.7.6])**.** *The let $f : X \to S$ be a morphism of simplicial sets. The following are equivalent.*

(i) f is a trivial Kan fibration;
(ii) f is a homotopy equivalence;
(iii) f is a weak homotopy equivalence;
(iv) for every vertex $s \in S$, the fiber $X_s = X \times_S \{s\}$ is contractible.

The next four types of fibration are all defined by the presence of various lifting properties with respect to various combinations of outer horn inclusions.

Definition 2.1.4 ([Lur09, 2.0.0.1, 2.0.0.3])**.** Let $p : X \to S$ be a morphism of simplicial sets. Then if p satisfies the right lifting property

$$\begin{tikzcd} \Lambda^n_k \arrow[r, "f"] \arrow[d, hook] & X \arrow[d, "p"] \\ \Delta^n \arrow[r, "\bar{f}"'] \arrow[ur, dashed] & S \end{tikzcd}$$

for every n, maps f and $\bar{f}$, and

(i) for $0 \leq k \leq n$, then p is a *Kan fibration*;
(ii) for $0 \leq k < n$, then p is a *left fibration*;
(iii) for $0 < k \leq n$, then p is a *right fibration*;
(iv) for $0 < k < n$, then p is an *inner fibration*.

Remark 2.1.5. It is easy to check that the classes of fibrations given in Definition 2.1.2 and Definition 2.1.4 are stable under pullback, as is any class of morphisms defined by a right lifting property.

Remark 2.1.6. A good way to motivate each of the types of fibrations above is to understand what can be said about their fibers. Let $p : X \to S$ be a map of simplicial sets, and $s : * \to S$ a vertex of S. We form the fiber $X_s = X \times_S \{s\}$. Then we have the following table of implications

Type of fibration	Structure on fiber
Inner fibration	∞-Category (Remark 2.2.1)
Left, right, or Kan fibration	Kan complex (2.4.4)
Trivial fibration	Contractible Kan complex (Proposition 2.1.3).

Example 2.1.7 (Left fibration which is not a Kan fibration)**.** Note that, by definition, a left fibration is a Kan fibration precisely if it is also a right fibration. It follows that the right horn inclusion $\Lambda^2_2 \hookrightarrow \Delta^2$, which is trivially a left fibration because it does not admit any nondegenerate left horn inclusions in dimension greater than 1, is not a Kan fibration.

Finally, we make the following group of definitions.

Definition 2.1.8 ([Lur09, 2.0.0.3])**.** Let $i : A \to B$ be a morphism of simplicial sets satisfying the left lifting property with respect to a class $\mathcal{F}$ of morphisms of simplicial sets. Then i is

(i) *anodyne* if $\mathcal{F}$ is the class of Kan fibrations;
(ii) *left anodyne* if $\mathcal{F}$ is the class of left fibrations;
(iii) *right anodyne* if $\mathcal{F}$ is the class of right fibrations; and
(iv) *inner anodyne* if $\mathcal{F}$ is the class of inner fibrations.

Example 2.1.9. By definition, The inclusion $\Lambda^n_k \hookrightarrow \Delta^n$ for $0 \leq k < n$ of any left horn is left anodyne.

2.2. More on inner fibrations and correspondences of ∞-categories

A primary motivation for studying inner fibrations is the following.

Remark 2.2.1. Note that Definition 1.1.4 can now be rephrased as follows; as simplicial set $\mathcal{C}$ is an ∞-category precisely if the unique map $\mathcal{C} \to *$ is an inner fibration. This is because the diagram Definition 1.1.4 is equivalent to

$$\begin{tikzcd} \Lambda^n_i \arrow[r] \arrow[d, hook] & \mathcal{C} \arrow[d] \\ \Delta^n \arrow[r] \arrow[ur, dashed] & *. \end{tikzcd}$$

Remark 2.2.2. A consequence of the previous remark is the following. Since the class of inner fibrations is stable under pullback, given any inner fibration $p : X \to S$, the fiber over an object s of S, which we denote X_s is automatically an ∞-category:

$$\begin{tikzcd} X_s \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & X \arrow[d, "p"] \\ * \arrow[r, "s"] & S. \end{tikzcd}$$

In this sense, we can think of X as a "family of ∞-categories parameterized by S."

Proposition 2.2.3 ([Lur25, 4.1.1.9])**.** *Let $p : X \to S$ be an inner fibration. If S is an ∞-category, then X is an ∞-category.*

However, the association $s \mapsto X_s$ is not necessarily functorial, i.e., there is in general no natural functor (of ∞-categories) $X_s \to X_{s'}$ associated to an edge $s \to s'$ in S. To be precise, we introduce the notion of a correspondence of ∞-categories.

Definition 2.2.4 ([Lur09, 2.3.1.3])**.** Let $\mathcal{C}$ and $\mathcal{C}'$ be ∞-categories. A *correspondence* between $\mathcal{C}$ and $\mathcal{C}'$ is an ∞-category $\mathcal{M}$ equipped with a functor $F : \mathcal{M} \to \Delta^1$ and equivalences $\mathcal{C} \simeq \mathcal{M} \times_\Delta \{0\}$ and $\mathcal{C}' \simeq \mathcal{M} \times_\Delta \{1\}$.

Remark 2.2.5. This definition is motivated by a more concrete notion of correspondences of ordinary categories. Such correspondences of ordinary categories $\mathcal{C}$ and $\mathcal{C}'$ are defined to be functors

$$M : \mathcal{C}^{op} \times \mathcal{C}' \to \mathsf{Set}\,.$$

One can show that this is equivalent to the data of a functor

$$F : \mathcal{M} \to [1],$$

where $\mathcal{M} = \mathcal{C} \star^M \mathcal{C}'$ is the category with all the objects and morphisms of $\mathcal{C}$ and $\mathcal{C}'$ and an additional morphism $c \to c'$ for every element of $M(c, c')$, and F is defined in the obvious way. This latter notion inspires the ∞-categorical Definition 2.2.4. See [Lur09, 2.3.1].

Remark 2.2.6 ([Lur09, 2.3.1.4])**.** There is a canonical way to go back and forth between (equivalence classes of) correspondences $F : \mathcal{M} \to \Delta^1$ between ∞-categories $\mathcal{C}$ and $\mathcal{C}'$ and functors $M : \mathcal{C}^{\mathsf{op}} \times \mathcal{C}' \to \mathcal{S}$. We won't go into detail about this because we don't need it, but roughly, given such an F, we can define M up to equivalence by

$$M(c, c') = \mathrm{Hom}_{\mathcal{M}}(c, c').$$

Construction 2.2.7. Also note that a functor $f : \mathcal{C}' \to \mathcal{C}$ of ∞-categories determines a correspondence $\mathcal{M} \to \Delta^1$ where the objects of $\mathcal{M}$ are the objects of $\mathcal{C}$ and $\mathcal{C}'$, and

$$\mathrm{Hom}_{\mathcal{M}}(x, y) = \begin{cases} \mathrm{Hom}_{\mathcal{C}}(x, y) & x, y \in \mathcal{C} \\ \mathrm{Hom}_{\mathcal{C}'}(x, y) & x, y \in \mathcal{C}' \\ \mathrm{Hom}_{\mathcal{C}'}(x, f(y)) & x \in \mathcal{C}, y \in \mathcal{C}' \\ \emptyset & x \in \mathcal{C}', y \in \mathcal{C}, \end{cases}$$

and the functor $\mathcal{M} \to \Delta^1$ is defined in the obvious way.

Note that composition in $\mathcal{M}$ makes sense exactly because f is a functor. That is, if $x \in \mathcal{C}$ and $y, z \in \mathcal{C}'$, then the composition in $\mathcal{M}$ is defined as follows:

$$\begin{tikzcd}
\mathrm{Hom}_{\mathcal{M}}(x,y) \times \mathrm{Hom}_{\mathcal{M}}(y,z) \arrow[r, "\circ_{\mathcal{M}}"] \arrow[d, equal] & \mathrm{Hom}_{\mathcal{M}}(x,z) \arrow[d, equal] \\
\mathrm{Hom}_{\mathcal{C}}(x,f(y)) \times \mathrm{Hom}_{\mathcal{C}'}(y,z) \arrow[r, dashed] \arrow[d, "1\times f"'] & \mathrm{Hom}_{\mathcal{C}}(x,f(z)) \\
\mathrm{Hom}_{\mathcal{C}}(x,f(y)) \times \mathrm{Hom}_{\mathcal{C}}(f(y),f(z)) \arrow[ur, "\circ_{\mathcal{C}}"'] &
\end{tikzcd}$$

The most important case for us will be the correspondence associated to an inner fibration.

Example 2.2.8. Let $p : X \to S$ be an inner fibration, and let $s \to s'$ be an edge of S. Then there is a correspondence $\mathcal{M} \to \Delta^1$ between the infinity categories X_s and $X_{s'}$ with the objects of X_s and $X_{s'}$ and morphisms

$$\mathrm{Hom}_{\mathcal{M}}(c,d) = \begin{cases} \mathrm{Hom}_{X_s}(c,d) & c,d \in X_s \\ \mathrm{Hom}_{\mathcal{C}'} & c,d \in X_{s'} \\ p^{-1}(\sigma : s \to s') & x \in X_s,\ y \in X_{s'} \\ \emptyset & c \in X_{s'},\ d \in X_s. \end{cases}$$

Note that the association from an edge $\sigma : s \to s'$ of S to the correspondence $\mathcal{M}$ is strictly weaker than associating a functor $f : X_{s'} \to X_s$. That is, Construction 2.2.7 does not generally have an inverse construction. Indeed, there is generally no natural choice of such a functor F, as we illustrate below.

Example 2.2.9. Let X be the simplicial set and $p : X \to \Delta^1$ be the inner fibration as in the following diagram:

$$\begin{tikzcd}
 & a \arrow[dr, "f"] & \\
X : & b \arrow[r, "g"] \arrow[d, dashed, "p"] & c \\
\Delta^1 : & 0 \arrow[r, "\sigma"] & 1.
\end{tikzcd}$$

The edge σ in Δ^1 determines a correspondence $\mathcal{M} \to \Delta^1$ where X_0 has objects a and b, X_1 has c as an object, and the only nontrivial mapping simplicial sets in $\mathcal{M}$ are

$$\mathrm{Hom}_{\mathcal{M}}(a,c) = \{f\},$$
$$\mathrm{Hom}_{\mathcal{M}}(b,c) = \{g\}.$$

If we wanted to associate $\mathcal{M} \to \Delta^1$ to a functor $\sigma^* : X_1 \to X_0$, then we would choose some edge $\tilde{\sigma} : x \to c$ in X with $p(\tilde{\sigma}) = s$, and then set $\sigma^*(c) = x$.

The issue here is that there are two inequivalent choices for the lift $\tilde{\sigma}$, namely f and g, which lead to inequivalent functors $\sigma^* : X_1 \to X_0$ sending c to a or b, respectively (and a and b are not isomorphic as objects of X_0). Thus, there is no *natural* choice of functor associated to the correspondence $\mathcal{M} \to \Delta^1$.

It turns out that a 1-categorical correspondence $M : \mathcal{C}^{\mathsf{op}} \times \mathcal{C}' \to \mathsf{Set}$ (as in Remark 2.2.5) arises from a functor $f : \mathcal{C}' \to \mathcal{C}$ via Construction 2.2.7 precisely if for every $c' \in \mathcal{C}'$, the functor

$$M(-,c') : \mathcal{C}^{\mathsf{op}} \to \mathsf{Set}$$

is representable by an object c of $\mathcal{C}$. See the discussion at the beginning of [Lur09, 2.4.1].

To obtain such a functor in the ∞-categorical case, we need to start with a more structured type of fibration $p : X \to S$. Specifically, we want a well-defined (up to equivalence) lift $\tilde{\sigma}$ of each edge σ in S. This leads us to the discussion of Cartesian fibrations in the next section.

Remark 2.2.10. It turns out that correspondences play a central role in the modern theory of ∞-categorical six functor formalisms. See [Sch26].

2.3. **Cartesian and coCartesian fibrations**

Let X be a simplicial set, and let $f : x \to y$ be an edge in X. Define $\mathsf{dom} : X_{/f} \to X_{/y}$ to be the canonical map of simplicial sets given on 2-simplices by

$$\begin{tikzcd} & a \arrow[dl] \arrow[dr, "g"] & \\ x \arrow[rr, "f"'] & & y \end{tikzcd} \quad \mapsto \quad \begin{tikzcd} a \arrow[d, "g"] \\ y. \end{tikzcd}$$

If X is an ∞-category, then the 2-simplex on the left can be thought of as a point of $\mathrm{Hom}_{X/y}(g, f)$, so that dom is the canonical map to $X_{/y}$ "taking the domain."

Definition 2.3.1 ([Lur09, 2.4.1.1])**.** Let $p : X \to S$ be an inner fibration of simplicial sets and $f : x \to y$ an edge in X. Then f is *p-Cartesian* if the induced map

$$X_{/f} \to X_{/y} \times_{S_{/p(y)}} S_{/p(f)}$$

as in the following diagram

$$\begin{tikzcd} X_{/f} \arrow[dr, dashed] \arrow[drr, bend left, "p"] \arrow[ddr, bend right, "\mathsf{dom}"'] & & \\ & X_{/y} \times_{S_{/p(y)}} S_{/p(f)} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & S_{/p(f)} \arrow[d, "\mathsf{dom}"] \\ & X_{/y} \arrow[r, "p"'] & S_{/p(y)} \end{tikzcd}$$

is a trivial Kan fibration (i.e., a homotopy equivalence).

Remark 2.3.2 ([Lur09, 2.4.1.4])**.** If $p : X \to S$ is an inner fibration, then an edge $f : x \to y$ in X is p-Cartesian if

$$X_{/f} \to X_{/y} \times_{S_{/p(y)}} S_{/p(f)}$$

admits a homotopy inverse. That is, given an edge $g : z \to y$ in X and a 2-simplex between the edges $p(g)$ and $p(f)$ in S, there exists some edge $h : x \to z$ in X making the following diagram commute up to higher coherence:

$$\begin{tikzcd} & z \arrow[dl, dashed, "h"'] \arrow[dr, "g"] & \\ x \arrow[rr, "f"'] \arrow[d, dashed, "p"] & & y \\ p(z) \arrow[dl] \arrow[dr, "p(g)"] & & \\ p(x) \arrow[rr, "p(f)"'] & & p(y). \end{tikzcd}$$

This is equivalent to a lift existing making the following diagram commute for all $n \geq 2$.

$$\begin{tikzcd} \Delta^{\{n-1,n\}} \arrow[d, hook] \arrow[dr, "f"] & & \\ \Lambda^n_n \arrow[r] \arrow[d, hook] & X \arrow[d, "p"] \\ \Delta^n \arrow[r] \arrow[ur, dashed] & S. \end{tikzcd}$$

Remark 2.3.3. Above, we have given a definition of p-Cartesian edges in the setting of general simplicial sets. This will be useful to us eventually, but a convenient (and relatively simple) alternate characterization of p-Cartesian edges in ∞-categories in terms of mapping spaces is given in Proposition 2.4.9.

Definition 2.3.4 ([Lur09, 2.4.2.1])**.** An inner fibration $p : X \to S$ is a *Cartesian fibration* if for every edge $\sigma : s_0 \to s_1$ of S and every vertex y of X with $p(y) = s_1$, there exists a p-Cartesian edge $f : x \to y$ of X with $p(x) = s_0$.

Remark 2.3.5. We can think of a Cartesian fibration as determining a "pullback square" for every edge $\sigma : s_0 \to s_1$ in S and $y \in p^{-1}(s_1)$:

$$\begin{tikzcd} \exists x \arrow[r, dashed, "\exists f"] \arrow[d, dashed, "p"'] & y \arrow[d, "p"] \\ s_0 \arrow[r, "\sigma"] & s_1. \end{tikzcd}$$

This is of course the motivation for the name "Cartesian." It is worth checking of course that the lifts x and f are canonically defined. Indeed this follows from the definitions; any other p-Cartesian edge $f' : x' \to y$ fitting into the above square is naturally equivalent to f by [Lur09, 2.4.1.5, 2.4.1.7].

Remark 2.3.6. Thus, Cartesian fibrations $p : X \to S$ can be thought of as precisely those maps of simplicial sets which have a "unique path lifting property" in the sense of algebraic topology. That is, given an edge $\sigma : s_0 \to s_1$ in the base S and an "endpoint" y lifting y, there is a lift $f : x \to y$ of σ through p *determined up to coherent higher homotopy.*

The following dual definitions will be useful to us as well.

Definition 2.3.7 ([Lur09, 2.4.1])**.** Let $p : X \to S$ be an inner fibration. An edge $f : x \to y$ of X is *p-coCartesian* if $f^{\mathsf{op}} : y \to x$ is Cartesian with respect to the inner fibration $p^{\mathsf{op}} : X^{\mathsf{op}} \to S^{\mathsf{op}}$.

Remark 2.3.8. Again, see Proposition 2.4.10 for a less opaque reformulation of this definition in the setting of ∞-categories.

Definition 2.3.9. An inner fibration $p : X \to S$ is a *coCartesian fibration* if $p^{\mathsf{op}} : X^{\mathsf{op}} \to S^{\mathsf{op}}$ is a Cartesian fibration.

That is, if for every edge $\sigma : s_0 \to s_1$ in S and vertex x in $p^{-1}(s_0)$, there is a p-coCartesian edge $f : x \to y$ in X with $p(f) = \sigma$, as in the following diagram:

$$\begin{tikzcd} x \arrow[r, dashed, "f"] \arrow[d, "p"'] & y \arrow[d, dashed, "p"] \\ s_0 \arrow[r, "\sigma"] & s_1 \end{tikzcd}$$

Definition 2.3.10. Let $\mathcal{D}$ be an ∞-category. We define $\mathsf{Cart}(\mathcal{D})$ (respectively, $\mathsf{coCart}(\mathcal{D})$) to be the subcategory of $(\mathsf{Cat}_\infty)_{/\mathcal{D}}$ with (co)Cartesian fibrations into $\mathcal{D}$ as objects and morphisms comprising the functors $F : \mathcal{C} \to \mathcal{C}'$ over $\mathcal{D}$ which send (co)Cartesian morphisms in $\mathcal{C}$ to (co)Cartesian morphisms in $\mathcal{C}'$.

Proposition 2.3.11 ([Lur09, 2.4.2.4])**.** *Let $p : X \to S$ be an inner fibration. The following are equivalent:*

(i) p is a Cartesian fibration, and every edge in X is p-Cartesian;
(ii) p is a Cartesian fibration, and every fiber of p is a Kan complex;
(iii) p is a right fibration.

It is convenient to introduce the following weaker notions of "locally Cartesian edges" and "locally Cartesian fibrations." These notions are generally easier to check, and with various additional conditions can be strengthened to honest Cartesian edges or fibrations.

Definition 2.3.12 ([Lur09, 2.4.1.11])**.** Let $p : X \to S$ be an inner fibration, and $e : \Delta^1 \to X$. We say e is a *locally p-Cartesian* edge of X if the natural map

$$e \times \mathsf{id} : \Delta^1 \to X \times_S \Delta^1$$

is a pr_2-Cartesian edge of $X \times_S \Delta^1$, where $\mathrm{pr}_2 : X \times_S \Delta^1 \to \Delta^1$ is the projection map.

Suppose e is an edge between vertices x and y of X. In the above definition, $X \times_S \Delta^1$ can be thought of as the collection of all edges in X lifting $p(e)$. Then e is locally p-Cartesian if it is the essentially unique lift of $p(e)$ with target y up to coherent homotopy.

Definition 2.3.13 ([Lur09, 2.4.2.6])**.** An inner fibration $p : X \to S$ is a *locally Cartesian fibration* if for each edge $\sigma : \Delta^1 \to S$, the pullback $X \times_S \Delta^1 \to \Delta^1$ is a Cartesian fibration.

A slight strengthening of the above definition recovers the notion of Cartesian fibration.

Proposition 2.3.14 ([Lur09, 2.4.2.10])**.** *An inner fibration $p : X \to S$ is a Cartesian fibration if for each $\sigma : \Delta^n \to S$ with $n = 0, 1, 2$, the pullback $X \times_S \Delta^n \to \Delta^n$ is a Cartesian fibration.*

2.4. Digression: mapping spaces

In this section, we record a few facts about mapping spaces in ∞-categories, as well as a few results which will serve as sources of left fibrations.

Proposition 2.4.1 ([Lur09, 2.1.2.2])**.** *Let $\mathcal{C}$ be an ∞-category, and $D : K \to \mathcal{C}$ any diagram in $\mathcal{C}$. The projection $\mathcal{C}_{D/} \to \mathcal{C}$ is a left fibration.*

Remark 2.4.2. Combining the above proposition with Remark 2.2.1, we note that $\mathcal{C}_{D/}$ itself is an ∞-category.

An application of this is the following.

Example 2.4.3. Given any object $c \in \mathcal{C}$, the natural projection map $p : \mathcal{C}_{c/} \to \mathcal{C}$ is a left fibration. In particular, given an object d of $\mathcal{C}$, we have a pullback square of ∞-categories

$$\begin{tikzcd} \mathsf{Maps}_{\mathcal{C}}(c,d) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{C}_{c/} \arrow[d, "p"] \\ * \arrow[r, "d"'] & \mathcal{C}. \end{tikzcd} \tag{2.4.4}$$

Since left fibrations are stable under pullback, it follows that $\mathsf{Maps}_{\mathcal{C}}(c,d) \to *$ is a left fibration, and so $\mathsf{Maps}_{\mathcal{C}}(c,d)$ is a Kan complex by Proposition 1.1.6.

Warning 2.4.5. In the previous example and proposition, we (somewhat abusively) used $\mathsf{Maps}_{\mathcal{C}}(c,d)$ where Lurie more accurately uses the simplicial set $\mathrm{Hom}^L(c,d)$. (See discussion at the beginning of [Lur09, 2.1.2]. For general definitions of the simplicial sets $\mathrm{Hom}^L(c,d)$ and $\mathrm{Hom}^R(c,d)$, see [Lur09, 1.2.2].) When $\mathcal{C}$ is an ∞-category, these objects are canonically equivalent as objects of $\mathsf{h}\mathcal{S}$.

This takes some work to prove, but the end result is that, if we don't need a *specific* simplicial model for $\mathsf{Maps}_{\mathcal{C}}(c,d)$ (i.e., we are only interested in its homotopy type), then it is perfectly fine to interpret (2.4.4) as a homotopy pullback in Set_Δ with the Joyal model structure (see Theorem 3.3.4) and take it as a definition of $\mathsf{Maps}_{\mathcal{C}}(c,d)$ as an object of $\mathsf{h}\mathcal{S}$. On the other hand, if we want (2.4.4) to be a categorical (or "strict") pullback in Set_Δ, then we must choose a specific model for $\mathsf{Maps}_{\mathcal{C}}(c,d)$, and the correct choice is $\mathrm{Hom}^L_{\mathcal{C}}(c,d)$.

Proposition 2.4.6 ([Lur09, 2.4.4.1])**.** *Let $p : \mathcal{C} \to \mathcal{C}'$ be an inner fibration of ∞-categories and let $c, d \in \mathcal{C}$. The induced map*

$$\mathsf{Maps}_{\mathcal{C}}(c,d) \to \mathsf{Maps}_{\mathcal{C}'}(p(c), p(d)) \tag{2.4.7}$$

is a Kan fibration.

Remark 2.4.8 ([Lur09, 2.4.4.2])**.** Under nice circumstances, we even have control over the homotopy fiber of the map in (2.4.7) over a vertex $\bar{e} : p(c) \to p(d)$. Specifically, if $\bar{e}$ admits a "locally p-Cartesian lift" (see Definition 2.3.12) $e : c' \to d$, then there is a homotopy Cartesian diagram

$$\begin{tikzcd} \mathsf{Maps}_{\mathcal{C}_{p(c)}}(c, c_0) \arrow[r, "e_*"] \arrow[d] & \mathsf{Maps}_{\mathcal{C}}(c,d) \arrow[d, "p"] \\ * \arrow[r, "\bar{e}"] & \mathsf{Maps}_{\mathcal{C}'}(p(c), p(d)) \end{tikzcd}$$

where $\mathcal{C}_{p(c)}$ is the fiber of p over $p(c)$ (which is an ∞-category by Remark 2.2.2).

The local condition in the previous remark assembles into the following nice result.

Proposition 2.4.9 ([Lur09, 2.4.4.3])**.** *Let $p : \mathcal{C} \to \mathcal{D}$ be an inner fibration of ∞-categories, and let $f : y \to z$ be a morphism in $\mathcal{C}$. Then f is p-Cartesian if and only if for every x in $\mathcal{C}$, the following diagram is homotopy Cartesian:*

$$\begin{tikzcd}
\mathsf{Maps}_{\mathcal{C}}(x,y) \arrow[r, "f_*"] \arrow[d, "p"'] & \mathsf{Maps}_{\mathcal{C}}(x,z) \arrow[d, "p"] \\
\mathsf{Maps}_{\mathcal{D}}(p(x),p(y)) \arrow[r, "p(f)_*"] & \mathsf{Maps}_{\mathcal{D}}(p(x),p(z)).
\end{tikzcd}$$

A dual version of the previous proposition gives us a convenient characterization of coCartesian morphisms.

Proposition 2.4.10. *Let $p : \mathcal{C} \to \mathcal{D}$ be an inner fibration of ∞-categories, and let $f : x \to y$ be a morphism in $\mathcal{C}$. Then f is p-coCartesian if and only if for every z in $\mathcal{C}$, the following diagram is homotopy Cartesian:*

$$\begin{tikzcd}
\mathsf{Maps}_{\mathcal{C}}(y,z) \arrow[r, "f^*"] \arrow[d, "p"'] & \mathsf{Maps}_{\mathcal{C}}(x,z) \arrow[d, "p"] \\
\mathsf{Maps}_{\mathcal{D}}(p(y),p(z)) \arrow[r, "p(f)^*"] & \mathsf{Maps}_{\mathcal{D}}(p(x),p(z)).
\end{tikzcd}$$

One more important source of left fibration is the following one regarding exponential objects.

Proposition 2.4.11 ([Lur09, 2.1.2.9])**.** *Let $p : X \to S$ be a left fibration and $i : A \to B$ a cofibration (i.e., monomorphism; Theorem 3.3.4) of simplicial sets. Then the natural map q obtained in the following diagram is a left fibration.*

$$\begin{tikzcd}
X^B \arrow[dr, dashed, "q"] \arrow[drr, bend left, "p_*"] \arrow[ddr, bend right, "i^*"'] & & \\
& X^A \times_{S^A} S^B \arrow[r] \arrow[d] & S^B \arrow[d, "i^*"] \\
& X^A \arrow[r, "p_*"'] & S^A
\end{tikzcd}$$

Moreover, if i is left anodyne, then q is a trivial Kan fibration.

3. Model structures

This diversity of fibrations outlined above leads to an array of various model structures on Set_Δ and related categories which we will discuss in this section. For convenience, we list them here:

(i) the Kan model structure on Set_Δ;
(ii) the Joyal model structure on Set_Δ;
(iii) a model structure on Cat_Δ;
(iv) the covariant and contravariant model structures on $(\mathsf{Set}_\Delta)_{/S}$; and
(v) the Cartesian and coCartesian model structures on $(\mathsf{Set}_\Delta^+)_{/S}$.

Many of these model structures are very closely related. In fact, Theorem 3.3.10 states that (ii) and (iii) are Quillen equivalent, and Theorem 3.4.10 provides Quillen adjunctions between (i), (ii), (iv), and (v) which in nice cases are also equivalences.

Many of these model structures come from the following abstract constructions which we describe in the second section:

(i) the combinatorial model structure specified by appropriate classes of cofibrations and weak equivalences; and
(ii) the projective and injective model structures on $\mathsf{Fun}(\mathcal{C}, \mathcal{A})$ induced by a model structure on $\mathcal{A}$.

First, we give a brief review of some relevant model category terminology.

3.1. **Recollections on model categories**

Definition 3.1.1 ([Lur09, A.2.4.1])**.** A model category $\mathcal{C}$ is *left proper* if for any cofibration $i : a \to b$ and weak equivalence $w : a \to a'$, the map $w' : b \to b'$ obtained in the following pushout square

$$\begin{tikzcd} a \arrow[r, "i"] \arrow[d, "w"'] \arrow[dr, phantom, "\ulcorner", very near start] & b \arrow[d, "w'"] \\ a' \arrow[r] & b' \end{tikzcd}$$

is also a weak equivalence. Dually, a definition of a *right proper* model category can be made with fibrations and pullback squares.

Proposition 3.1.2 ([Lur09, A.2.4.2])**.** *Let $\mathcal{C}$ be a model category in which every object is cofibrant. Then $\mathcal{C}$ is left proper.*

Recall that a morphism in a model category $\mathcal{C}$ is called a *trivial (or acyclic) cofibration* if it is both a cofibration and a weak equivalence and a *trivial (or acyclic) fibration* if it is both a fibration and a weak equivalence. An object $c \in \mathcal{C}$ is *cofibrant* if the unique map $\emptyset \to c$ is a cofibration, and *fibrant* if the unique map $c \to *$ is a fibration.

Definition 3.1.3 ([Lur09, A.2.5])**.** Let $\mathcal{C}$ be a model category. Let $\mathcal{C}_{\mathrm{cof}}$ be the full subcategory spanned by the cofibrant objects, and let W be the class of weak equivalences in $\mathcal{C}_{\mathrm{cof}}$. The *homotopy category* of $\mathcal{C}$ is the localization of $\mathcal{C}_{\mathrm{cof}}$ at W:

$$\mathsf{h}\mathcal{C} := \mathcal{C}_{\mathrm{cof}}[W^{-1}].$$

Note that the homotopy category $\mathsf{h}\mathcal{C}$ is an ordinary category, no longer equipped with a model structure.

Definition 3.1.4 ([Lur09, A.2.5])**.** Let $\mathcal{C}$ and $\mathcal{D}$ be model categories, and let

$$\begin{tikzcd} \mathcal{C} \arrow[r, shift left, "F"] & \mathcal{D} \arrow[l, shift left, "G"] \end{tikzcd}$$

be a pair of adjoint functors. We say $F \dashv G$ is a *Quillen adjunction* if any of the following equivalent conditions hold:

(i) F preserves cofibrations and trivial cofibrations;
(ii) G preserves fibrations and trivial fibrations;
(iii) F preserves cofibrations and G preserves fibrations;
(iv) F preserves trivial cofibrations and G preserves trivial fibrations.

One can check that, given a Quillen adjunction $F \dashv G$ as above, there are natural induced adjoint functors $LF \dashv RG$ on homotopy categories called the *left derived functor* of F and the *right derived functor* of G, respectively:

$$\begin{tikzcd} \mathsf{h}\mathcal{C} \arrow[r, shift left, "LF"] & \mathsf{h}\mathcal{D}. \arrow[l, shift left, "RG"] \end{tikzcd}$$

Definition 3.1.5 ([Lur09, A.2.5])**.** A Quillen adjunction $F \dashv G$ is a *Quillen equivalence* if any of the following equivalent conditions hold:

(i) $LF : \mathsf{h}\mathcal{C} \to \mathsf{h}\mathcal{D}$ is an equivalence of categories;
(ii) $RG : \mathsf{h}\mathcal{D} \to \mathsf{h}\mathcal{C}$ is an equivalence of categories;
(iii) for every cofibrant $c \in \mathcal{C}$ and fibrant $d \in \mathcal{D}$, a morphism $c \to G(d)$ is a weak equivalence if and only if its adjoint $F(c) \to d$ is.

There is an elaborate theory of *enriched model categories* described in [Lur09, A.3.1]. For our purposes, we will be most interested in the case where our model categories are enriched over Set_Δ. We record the following definition.

Definition 3.1.6 ([Lur09, A.3.1.5])**.** A *simplicial model category* is a Set_Δ-enriched category $\mathcal{C}$ which is

(i) *tensored* over Set_Δ; i.e., there exists a functor $- \otimes - : \mathsf{Set}_\Delta \times \mathcal{C} \to \mathcal{C}$ and a natural isomorphism

$$\mathrm{Hom}_{\mathcal{C}}(X \otimes c, c') \cong \mathrm{Hom}_{\mathsf{Set}_\Delta}(X, \mathrm{Hom}_{\mathcal{C}}(c, c'));$$

(ii) *cotensored* over Set_Δ; i.e., there exists a functor $[-,-] : (\mathsf{Set}_\Delta)^{\mathsf{op}} \times \mathcal{C} \to \mathcal{C}$ and a natural isomorphism

$$\mathrm{Hom}_{\mathcal{C}}(c, [X, c']) \cong \mathrm{Hom}_{\mathsf{Set}_\Delta}(X, \mathrm{Hom}_{\mathcal{C}}(c, c'));$$

and which is equipped with a model structure compatible with $- \otimes -$ in an appropriate sense (see [Lur09, A.3.1.1]). In particular, $- \otimes -$ preserves colimits in both variables.

3.2. Some general constructions of model structures

For the next definition, recall that a class S of morphisms in a category $\mathcal{C}$ is *weakly saturated* if it is closed under pushouts, under transfinite compositions, and under retracts. (See [Lur09, A.1.2.2].) A locally small category $\mathcal{C}$ is called *presentable* if it admits all small colimits, is generated by a (small) set of objects under colimits, and every object is compact. (See [Lur09, A.1.1.2].)

Definition 3.2.1 ([Lur09, A.2.6.1])**.** Let $\mathcal{C}$ be a model category. We say $\mathcal{C}$ is *combinatorial* if it satisfies the following conditions:

(i) $\mathcal{C}$ is presentable;
(ii) there exists a set I of *generating cofibrations*, so that the class of cofibrations in $\mathcal{C}$ is the smallest weakly saturated class of morphisms containing I;
(iii) there exists a set J of *generating trivial cofibrations*, so that the class of cofibrations in $\mathcal{C}$ is the smallest weakly saturated class of morphisms containing J.

The upshot of the notion of combinatorial model categories is that they are relatively easy to produce. Indeed, we have the following.

Proposition 3.2.2 ([Lur09, A.2.6.15])**.** *Let $\mathcal{C}$ be a presentable category, and suppose W is a class of morphisms in $\mathcal{C}$ which we will call weak equivalences, and C_0 is any set of morphisms in $\mathcal{C}$. Suppose that*

(i) the class W is perfect*; i.e., W contains all isomorphisms, satisfies a "two-out-of-three" property with respect to compositions, is closed under filtered colimits, and is generated by a subset $W_0 \subset W$ under filtered colimits (see [Lur09, A.2.6.12]);*
(ii) given a diagram of two pushout squares

$$\begin{tikzcd} a \arrow[r] \arrow[d, "i_0"'] \arrow[dr, phantom, "\ulcorner"] & a' \arrow[r, "w"] \arrow[d] \arrow[dr, phantom, "\ulcorner"] & a'' \arrow[d] \\ b \arrow[r] & b' \arrow[r, "w'"] & b'' \end{tikzcd}$$

in which i_0 in C_0 and w in W, then w' is also in W; and
(iii) if g is a morphism in $\mathcal{C}$ satisfying the right lifting property with respect to W_0, then g is in W;

then there exists a left-proper combinatorial model structure on $\mathcal{C}$ in which a morphism $f : c \to d$ is

(C) a cofibration *if it belongs to the weakly saturated class generated by C_0;*
(W) a weak equivalence *if it belongs to W; and*
(F) a fibration *if it has the right lifting property with respect to the class of trivial cofibrations.*

A first application of the previous very powerful proposition is the construction of the following two model structures on functor categories.

Theorem 3.2.3 (Projective model structure on $\mathsf{Fun}(\mathcal{C}, \mathcal{A})$ [Lur09, A.2.8.1, A.2.8.2])**.** *Let $\mathcal{C}$ be a small category, and let $\mathcal{A}$ be a model category. There is a combinatorial model structure on $\mathsf{Fun}(\mathcal{C}, \mathcal{A})$ in which a morphism $\eta : F \to F'$ is*

(F) a projective fibration *if $F(c) \to G(c)$ is a fibration in $\mathcal{A}$ for each c in $\mathcal{C}$;*
(W) a weak equivalence *if $F(c) \to G(c)$ is a weak equivalence in $\mathcal{A}$ for each c in $\mathcal{C}$;*
(C) a projective cofibration *if it has the* left *lifting property with respect to every trivial fibration.*

Remark 3.2.4. The *injective model structure* on $\mathsf{Fun}(\mathcal{C}, \mathcal{A})$ is defined in a dual way by setting the *injective cofibrations* to be those morphisms which descend to cofibrations in $\mathcal{A}$ at every c in $\mathcal{C}$ and using the same weak equivalences.

Remark 3.2.5. The projective and injective model structures are essentially the two natural choices of model structures on $\mathsf{Fun}(\mathcal{C}, \mathcal{A})$ induced by the model structure on $\mathcal{A}$. In general they do not coincide.

3.3. Model structures on Set_Δ and Cat_Δ

We begin with the most elementary homotopy-theoretic model structure on Set_Δ.

Theorem 3.3.1 (Kan model structure on Set_Δ [Lur09, A.2.7])**.** *The category Set_Δ admits a (combinatorial, cofibrant, left proper, and right proper) model structure in which*

(C) the cofibrations *are the monomorphisms of simplicial sets, i.e., the maps $i : S \to S'$ inducing injections $S_n \to S'_n$ for each n (or, equivalently, those maps i satisfying the left lifting property with respect to every trivial fibration),*

(W) the weak equivalences *are the maps $w : X \to Y$ which induce homotopy equivalences $|X| \to |Y|$ on geometric realizations, and*

(F) the fibrations *are the Kan fibrations (Definition 2.1.4).*

Remark 3.3.2. The Kan model structure is sometimes also called the Quillen model structure, or even just the "standard" or "usual" model structure on Set_Δ.

Using the Kan model structure on the symmetric monoidal category Set_Δ, one can use Proposition 3.2.2 to define the following model structure on Cat_Δ.

Theorem 3.3.3 (Model structure on Cat_Δ [Lur09, A.3.2.2, A.3.2.3, A.3.2.4])**.** *The category Cat_Δ admits a left proper combinatorial model structure in which*

(C) the cofibrations *are comprise the smallest weakly saturated class of morphisms containing*

(i) the inclusion of categories $\emptyset \hookrightarrow *$

(ii) the natural functors $[1]_S \to [1]_{S'}$ induced by cofibrations $S \to S'$ of simplicial sets, where $[1]_S$ is the simplicial category with two objects a and b and

$$\mathrm{Hom}_{[1]_S}(x, y) = \begin{cases} S & x = a,\ y = b \\ \emptyset & x = b,\ y = a \\ * & x = y \end{cases}$$

(W) the weak equivalences *are the morphisms $F : \mathcal{C} \to \mathcal{D}$ of simplicial categories inducing an equivalence $hF : \mathsf{h}\mathcal{C} \to \mathsf{h}\mathcal{D}$ on homotopy categories.*

(F) the fibrations *are the morphisms with the right lifting property with respect to the class of trivial cofibrations.*

Now, we introduce an important alternate model structure on Set_Δ.

Theorem 3.3.4 (Joyal model structure on Set_Δ [Lur09, 2.2.5.1])**.** *There exists a left proper model structure on Set_Δ in which*

(C) the cofibrations *$i : S \to S'$ are the monomorphisms of simplicial sets (as in Theorem 3.3.1),*

(W) the categorical equivalences *$w : S \to S'$ are those maps of simplicial sets inducing an equivalence $\mathfrak{C}[S] \to \mathfrak{C}[S']$ of simplicial categories, and*

(F) the categorical fibrations *$p : S \to S'$ are those maps of simplicial sets with the right lifting property with respect to all trivial cofibrations.*

Remark 3.3.5. Note that the cofibrations of the Joyal model structure coincide with those of the Kan model structure (Theorem 3.3.1). In particular, the Joyal model structure is cofibrant (every object is cofibrant).

The terminology "categorical equivalence" and "categorical fibration" in Theorem 3.3.4, used instead of "weak equivalence" or simply "fibration," is to avoid ambiguity with, e.g., the weak equivalences of the Kan model structure [Lur09, 1.1.5.16].

Note that a categorical equivalence $w : S \to S'$ of simplicial sets is a map of simplicial sets which induces an equivalence of $\mathsf{h}\mathcal{S}$-enriched categories $Lw : \mathsf{h}S \to \mathsf{h}S'$, where the *homotopy category of a simplicial set S* is defined to be the (ordinary) category $\mathsf{h}S$ where the objects are the vertices of S, and

$$\mathrm{Hom}_{\mathsf{h}S}(x, y) = [\mathrm{Hom}_{\mathfrak{C}[S]}(x, y)].$$

At the level of mapping objects, this is a special case of the general definition of the homotopy category of a model category (Definition 3.1.3) applied to the Kan model structure on Set_Δ.

The following fact ties these various notions together.

Proposition 3.3.6 ([Lur09, 1.1.5.17])**.** *The following are equivalent:*

(i) $f : S \to S'$ *is a categorical equivalence of simplicial sets;*
(ii) $\mathfrak{C}[f] : \mathfrak{C}[S] \to \mathfrak{C}[S']$ *is an equivalence of simplicial categories;*
(iii) $|\mathfrak{C}[f]| : |\mathfrak{C}[S]| \to |\mathfrak{C}[S']|$ *is an equivalence of topological categories (in the sense of Definition 1.2.5).*

Remark 3.3.7. Any one of the three equivalent notions of "categorical equivalence" can fill the role of *equivalence of* ∞*-categories* , depending on the context. Note that we reserve the term *isomorphism of* ∞*-categories* for maps of ∞-categories which are isomorphisms of simplicial sets (this is generally a much stronger property).

Since every inner anodyne map is a categorical equivalence, any categorical fibration is an inner fibration [Lur09, 2.2.5.5]. It is not generally true that an inner fibration $p : S \to S'$ is a categorical fibration, but it is if $S' = *$. Thus, following Remark 2.2.1, we arrive at the following theorem.

Theorem 3.3.8 ([Lur09, 2.4.6.1])**.** *The fibrant objects in* Set_Δ *with respect to the Joyal model structure are precisely the* ∞*-categories.*

Remark 3.3.9. Since the Joyal model structure is cofibrant, we might sometimes refer to ∞-categories as the *bifibrant* (fibrant and cofibrant) objects of Set_Δ, the full subcategory of which we denote by $\mathsf{Set}_\Delta^\circ$.

An important takeaway of this section is the following.

Theorem 3.3.10 ([Lur09, 2.2.5.1])**.** *With respect the the model structures of Theorem 3.3.3 and Theorem 3.3.4, the adjunction*

$$\mathfrak{C}[-] : \mathsf{Set}_\Delta \rightleftarrows \mathsf{Cat}_\Delta : N$$

is a Quillen equivalence.

Thus, we can alternately make arguments about simplicial sets and simplicial categories depending on which is more convenient. The straightening and unstraightening constructions will make use of both.

3.4. Model structures on $(\mathsf{Set}_\Delta)_{/S}$ and a big comparison theorem

There are many model structures we can put on $(\mathsf{Set}_\Delta)_{/S}$, which we record in this section. See Theorem 3.4.10 for a summary of their relationships.

Theorem 3.4.1 (Covariant model structure on $(\mathsf{Set}_\Delta)_{/S}$ [Lur09, 2.1.4.5, 2.1.4.7])**.** *Let* S *be a simplicial set. There is a left proper combinatorial model structure on* $(\mathsf{Set}_\Delta)_{/S}$ *in which a morphism* $f : X \to Y$ *over* S *is*

(C) *a* covariant cofibration *if it is a monomorphism of simplicial sets;*
(W) *a* covariant equivalence *if the induced map*

$$(X^{\triangleleft}) \coprod_X S \to (Y^{\triangleleft}) \coprod_Y S$$

is a categorical equivalence in the sense of Theorem 3.3.4;
(F) *a* covariant fibration *if it has the right lifting property with respect to all trivial cofibrations.*

Remark 3.4.2. Note that the *(left) cone* $C^{\triangleleft}(f)$ of a morphism $f : X \to S$ of simplicial sets is defined in [Lur09, 2.1.4.2] to be the simplicial set

$$S \coprod_X (X^{\triangleleft}).$$

For technical reasons, this is backwards from the setup of covariant equivalences; nevertheless, a covariant equivalence can be roughly interpreted in a familiar way as a map over S inducing an equivalence on cones.

Proposition 3.4.3 ([Lur09, 2.2.3.12])**.** *The fibrant objects of* $(\mathsf{Set}_\Delta)_{/S}$ *with the covariant model structure are precisely the left fibrations* $X \to S$*.*

Remark 3.4.4. There is a *contravariant model structure* on $(\mathsf{Set}_\Delta)_{/S}$ described in [Lur09, 2.1.4.12] which is obtained from the covariant model structure by replacing f with f^{op}. An important result about the contravariant model structure is the following.

Proposition 3.4.5 ([Lur09, 2.2.3.12, 2.2.3.14])**.** *Let $p : X \to S$ be a map of simplicial sets. Then p is a right fibration if and only if it is fibrant with respect to the contravariant model structure on $(\mathsf{Set}_\Delta)_{/S}$.*

In fact, if $q : Y \to S$ is also a right fibration, then a morphism $f : X \to Y$ over S is a right fibration if and only if it is a contravariant fibration.

Next, we define a model structure on the related category $(\mathsf{Set}_\Delta^+)_{/S}$ of marked simplicial sets over S.

Theorem 3.4.6 (Cartesian model structure on $(\mathsf{Set}_\Delta^+)_{/S}$ [Lur09, 3.1.3.7])**.** *Let S be a simplicial set. There is a left proper combinatorial model structure on $(\mathsf{Set}_\Delta^+)_{/S}$ in which*

(C) the cofibrations *are the morphisms $i : X \to Y$ over S in which the underlying map of simplicial sets $X \to Y$ is a cofibration in the sense of Theorem 3.3.1;*

(W) the Cartesian equivalences *are the morphisms $w : X \to Y$ over S inducing an equivalence of ∞-categories*

$$\mathsf{Maps}_S^\flat(Y, Z^\natural) \to \mathsf{Maps}_S^\flat(X, Z^\natural)$$

for every Cartesian fibration $Z \to S$ (see Remark 1.4.5, [Lur09, 3.1.3.3]);

(F) the fibrations *are the morphisms with the right lifting property with respect to the trivial cofibrations.*

Remark 3.4.7. We will sometimes refer to the Cartesian model structure on Set_Δ^+, by which we really mean the Cartesian model structure on $(\mathsf{Set}_\Delta^+)_{/*}$.

Remark 3.4.8 ([Lur09, 3.1.3.9])**.** Dually, we can define a *coCartesian model structure* on $(\mathsf{Set}_\Delta^+)_{/S}$. These are in general not equivalent, but they are when S is a Kan complex [Lur09, 3.1.4.6].

Note that the notions of Cartesian fibrations of simplicial sets (Definition 2.3.4) and the fibrations in the Cartesian model structure (Theorem 3.4.6) are distinct, though closely related, as the following proposition shows.

Proposition 3.4.9 ([Lur09, 3.1.4.1])**.** *Let $X \in (\mathsf{Set}_\Delta^+)_{/S}$. The following are equivalent:*

(i) X is fibrant with respect to the Cartesian model structure;

(ii) there is a Cartesian fibration $Y \to S$ and an isomorphism $X \cong Y^\natural$ in $(\mathsf{Set}_\Delta^+)_{/S}$.

The following large theorem relates all the model structures discussed in this section.

Theorem 3.4.10 ([Lur09, 3.1.5.1])**.** *There is a sequence of Quillen adjunctions*

$$\mathcal{C}_{\text{Joyal}} \underset{U}{\overset{(-)^\flat}{\rightleftarrows}} \mathcal{C}_{\text{Cart}} \underset{\mathsf{id}}{\overset{\mathsf{id}}{\rightleftarrows}} \mathcal{C}'_{\text{Cart}} \underset{(-)^\sharp}{\overset{U}{\rightleftarrows}} \mathcal{C}_{\text{cont}} \underset{\mathsf{id}}{\overset{\mathsf{id}}{\rightleftarrows}} \mathcal{C}_{\text{Kan}}$$

in which U is the forgetful functor $(\mathsf{Set}_\Delta^+)_{/S} \to (\mathsf{Set}_\Delta)_{/S}$, and

$\mathcal{C}_{\text{Joyal}}$ *is $(\mathsf{Set}_\Delta)_{/S}$ with the Joyal model structure (Theorem 3.3.4);*

$\mathcal{C}_{\text{Cart}}$ *is $(\mathsf{Set}_\Delta^+)_{/S}$ with the Cartesian model structure (Theorem 3.4.6);*

$\mathcal{C}'_{\text{Cart}}$ *is $(\mathsf{Set}_\Delta^+)_{/S}$ with a localization of the Cartesian model structure in which a map $(X, \mathcal{E}_X) \to (Y, \mathcal{E}_Y)$ is a cofibration if $X \to Y$ is a monomorphism of simplicial sets, and a weak equivalence if $X^\sharp \to Y^\sharp$ is a Cartesian equivalence [Lur09, 3.1.5.2];*

$\mathcal{C}_{\text{cont}}$ *is $(\mathsf{Set}_\Delta)_{/S}$ with the contravariant model structure (Remark 3.4.4); and*

$\mathcal{C}_{\text{Kan}}$ *is $(\mathsf{Set}_\Delta)_{/S}$ with the Kan model structure (Theorem 3.3.1).*

Moreover, the first and fourth Quillen adjunctions are Quillen equivalences whenever S is a Kan complex, and the third adjunction is always a Quillen equivalence.

4. Straightening and unstraightening

4.1. The classical Grothendieck construction

Roughly, the classical Grothendieck construction establishes an equivalence between functors $\chi : \mathcal{D} \to \mathsf{Gpd}$ and functors $F : \mathcal{C} \to \mathcal{D}$ "cofibered in groupoids." We begin with a definition.

Definition 4.1.1 ([Lur09, 2.1.1.1])**.** Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. We say that $\mathcal{C}$ is *cofibered in groupoids (over $\mathcal{D}$)* if

(i) morphisms in $\mathcal{D}$ lift to morphisms in $\mathcal{C}$; that is, for every $c \in \mathcal{C}$ and morphism $\eta : F(c) \to d$ in $\mathcal{D}$, there exists a morphism $\tilde{\eta} : c \to \tilde{d}$ in $\mathcal{C}$ with $F(\tilde{\eta}) = \eta$ as in the following diagram

$$\begin{tikzcd} c \arrow[r, dashed, "\tilde{\eta}"] \arrow[d, mapsto, "F"'] & \tilde{d} \arrow[d, mapsto, "F"] \\ F(c) \arrow[r, "\eta"] & d \end{tikzcd}$$

(ii) for every morphism $\eta : c \to c'$ in $\mathcal{C}$ and object c'' in $\mathcal{C}$, the following diagram is Cartesian:

$$\begin{tikzcd} \mathrm{Hom}_{\mathcal{C}}(c', c'') \arrow[r, "\eta"] \arrow[d, "F"'] & \mathrm{Hom}_{\mathcal{C}}(c, c'') \arrow[d, "F"] \\ \mathrm{Hom}_{\mathcal{D}}(F(c'), F(c'')) \arrow[r, "F(\eta)^*"] & \mathrm{Hom}_{\mathcal{D}}(F(c), F(c'')) \end{tikzcd}$$

Remark 4.1.2. Point (ii) in Definition 4.1.1 can be rephrased in the following way. For any morphisms $\eta : c \to c'$ and $\alpha : c \to c''$ in $\mathcal{C}$ and $\beta : F(c') \to F(c'')$ in $\mathcal{D}$, there exists some (necessarily unique) morphism $\gamma : c' \to c''$ making the following diagram commute.

$$\begin{tikzcd} c \arrow[r, "\eta"] \arrow[rr, bend left, "\alpha"] \arrow[d, mapsto, "F"'] & c' \arrow[r, dashed, "\gamma"] \arrow[d, mapsto, "F"'] & c'' \arrow[d, mapsto, "F"'] \\ F(c) \arrow[r, "F(\eta)"] & F(c') \arrow[r, "\beta"] & F(c'') \end{tikzcd}$$

In particular, taking $c = c''$ and $\alpha = \mathsf{id}_c$, this implies that every morphism η admits a retract, from which it follows that every morphism in $\mathcal{C}$ is invertible, i.e., the fibers of F are in fact groupoids.

With this definition, we are equipped to recall the classical Grothendieck construction which establishes the following correspondence:

$$\left\{ \begin{array}{c} F : \mathcal{C} \to \mathcal{D} \\ \text{cofibered in groupoids} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{Functors} \\ \chi : \mathcal{D} \to \mathsf{Gpd} \end{array} \right\}.$$

The rest of this section is devoted to describing the backward and forward directions of this construction. One can check that these two constructions are canonically "inverse" to each other in some appropriate sense.

Construction 4.1.3 (Classical Grothendieck construction: unstraightening direction [Lur09, 2.1.1])**.** Let $\mathcal{D}$ be an (ordinary, small) category, and consider a functor

$$\chi : \mathcal{D} \to \mathsf{Gpd}\,.$$

We define a category $\mathcal{C}_\chi$ with objects consisting of pairs (d, η) where $d \in \mathcal{D}$ and η is an object of the groupoid $\chi(d)$. Given two objects (d, η) and $(d'\eta')$ of $\mathcal{C}_\chi$, a morphism from (d, η) to (d', η') in $\mathcal{C}_\chi$ is a pair (f, α) where $f : d \to d'$ is a morphism in $\mathcal{D}$ and $\alpha : \chi(f)(\eta) \to \eta'$ is an (iso)morphism in $\chi(d')$, as in the following diagram:

$$\begin{tikzcd} \eta \arrow[r, mapsto, "\chi(f)"] \arrow[d, mapsto, "F"'] & \chi(f)(\eta) \arrow[r, "\cong", "\alpha"'] \arrow[d, mapsto, "F"'] & \eta' \arrow[dl, mapsto, bend left, "F"] \\ d \arrow[r, "f"] & d' & \end{tikzcd}$$

Composition in $\mathcal{C}_\chi$ is defined in the obvious way, and one can check that, equipped with the evident forgetful functor to $\mathcal{D}$, $\mathcal{C}_\chi$ is cofibered in groupoids over $\mathcal{D}$.

Construction 4.1.4 (Classical Grothendieck construction: straightening direction [Lur09, 2.1.1])**.** Given a category cofibered in groupoids $F : \mathcal{C} \to \mathcal{D}$, we can obtain a functor $\chi : \mathcal{D} \to \mathsf{Gpd}$ in the following way. On an object d of $\mathcal{D}$ we define

$$\chi(d) = \mathcal{C}_d = \mathcal{C} \times_{\mathcal{D}} \{d\}$$

(which one can check is in fact a groupoid by Definition 4.1.1 (ii)). Notice that $\mathcal{C}_d$ can equivalently be described as the category of lifts of $d : * \to \mathcal{D}$ through F, as in the following diagram.

$$\begin{tikzcd} & \mathcal{C} \arrow[d, "F"] \\ * \arrow[ur, dashed, "c"] \arrow[r, "d"'] & \mathcal{D} \end{tikzcd} \tag{4.1.5}$$

Notice that from this construction it is clear that a morphism in $\mathcal{C}_d$ must cover id_d, and not some other automorphism of d.

Next, we let χ send a morphism $f : d \to d'$ in $\mathcal{D}$ to the functor $f_! : \mathcal{C}_d \to \mathcal{C}_{d'}$ defined in the following way. For each $c \in \mathcal{C}_d$, use Definition 4.1.1 (i) to obtain a morphism $\tilde{f} : c \to c'$ lifting f, and define $f_!(c) = c'$. Since each $\tilde{f}$ is defined up to canonical isomorphism, we obtain a functor $f_!$.

However, the associations $d \mapsto \mathcal{C}_d$ and $f \mapsto f_!$ do not assemble into an honest functor $\chi : \mathcal{D} \to \mathsf{Gpd}$ since the composition law $(f \circ g)_! = f_! \circ g_!$ holds only up to canonical isomorphism and is not necessarily a strict equality. This is remedied by replacing the definition of $\mathcal{C}_d$ in (4.1.5) with the category of lifts

$$\begin{tikzcd} & \mathcal{C} \arrow[d, "F"] \\ \mathcal{D}_{d/} \arrow[ur, dashed, "\ell"] \arrow[r, "\pi^!"] & \mathcal{D} \end{tikzcd}$$

i.e., the fiber of

$$F_* : \mathsf{Fun}(\mathcal{D}_{d/}, \mathcal{C}) \to \mathsf{Fun}(\mathcal{D}_{d/}, \mathcal{D})$$

over the forgetful functor $U : \mathcal{D}_{d/} \to \mathcal{D}$ which sends $d \to e$ to e. This fiber is canonically equivalent to $\mathcal{C} \times_{\mathcal{D}} \{d\}$. However, with this definition, a morphism $f : d \to d'$ in $\mathcal{D}$ determines a functor $f^* : \mathcal{D}_{d'/} \to \mathcal{D}_{d/}$ which yields

$$\begin{aligned} f^{**} : \mathsf{Fun}(\mathcal{D}_{d/}, \mathcal{C}) &\longrightarrow \mathsf{Fun}(\mathcal{D}_{d'/}, \mathcal{C}) \\ \ell &\longmapsto f^* \circ \ell. \end{aligned}$$

This functor lifts to maps on the fibers of F_* which satisfy a strict composition law. That is, for morphisms $f : d \to d'$ and $g : d' \to d''$ in $\mathcal{D}$, we obtain a diagram

$$\begin{tikzcd} \mathcal{C}_d \arrow[r, dashed, "f_!"] \arrow[rr, dashed, bend left, "(g\circ f)_!"] \arrow[d, "\mathrm{fib}(U)"'] & \mathcal{C}_{d'} \arrow[r, dashed, "g_!"] \arrow[d, "\mathrm{fib}(U)"'] & \mathcal{C}_{d''} \arrow[d, "\mathrm{fib}(U)"'] \\ \mathsf{Fun}(\mathcal{D}_{d/}, \mathcal{D}) \arrow[r, "f^{**}"] \arrow[d, "F_*"'] & \mathsf{Fun}(\mathcal{D}_{d'/}, \mathcal{D}) \arrow[r, "g^{**}"] \arrow[d, "F_*"'] & \mathsf{Fun}(\mathcal{D}_{d''/}, \mathcal{D}) \arrow[d, "F_*"'] \\ \mathsf{Fun}(\mathcal{D}_{d/}, \mathcal{C}) \arrow[r, "f^{**}"] \arrow[rr, bend right, "(g\circ f)^{**}"'] & \mathsf{Fun}(\mathcal{D}_{d'/}, \mathcal{C}) \arrow[r, "g^{**}"] & \mathsf{Fun}(\mathcal{D}_{d''/}, \mathcal{C}) \end{tikzcd}$$

which is strictly commutative (essentially because Cartesian squares "glue").

Remark 4.1.6. There is a dual version of the Grothendieck construction establishing a correspondence

$$\left\{ \begin{array}{c} F : \mathcal{C} \to \mathcal{D} \\ \text{fibered in groupoids} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{Functors} \\ \chi : \mathcal{D}^{\mathsf{op}} \to \mathsf{Gpd} \end{array} \right\}.$$

This version is somewhat more familiar in algebraic geometry. To avoid dealing with contravariant functors, we will prefer the version described above.

4.2. **Left fibrations and the Grothendieck construction**

One can ask what the correct ∞-categorical formulation of the Grothendieck construction is. It seems like our functors $\chi : \mathcal{D} \to \mathsf{Gpd}$ out of an ordinary category $\mathcal{D}$ might appropriately be replaced with functors $\chi : \mathcal{D} \to \mathcal{S}$ to the ∞-category of ∞-groupoids when $\mathcal{D}$ is an ∞-category. But how should we replace functors $F : \mathcal{C} \to \mathcal{D}$ cofibered in Gpd? It turns out the place to start is with left fibrations. Indeed, an initial result is the following.

Proposition 4.2.1 ([Lur09, 2.1.1.3])**.** *A functor $F : \mathcal{C} \to \mathcal{D}$ of ordinary categories is cofibered in groupoids if and only if its nerve $N(F) : N(\mathcal{C}) \to N(\mathcal{D})$ is a left fibration (of simplicial sets).*

Recall from (2.4.4) that the fibers of a left fibration $p : X \to S$ are Kan complexes (∞-groupoids), just as the fibers of $F : \mathcal{C}_\chi \to \mathcal{D}$ are (ordinary) groupoids in Construction 4.1.3. Moreover, given an edge $f : s \to s'$ in s, there exists a lift as in the following diagram

$$\begin{tikzcd} \{0\} \times X_s \arrow[r] \arrow[d, hook] & X \arrow[d, "p"] \\ \Delta^1 \times X_s \arrow[r, "\mathrm{pr}_1"'] \arrow[ur, dashed, "\ell"] & \Delta^1 \arrow[r, "f"] & S \end{tikzcd}$$

determining a map of simplicial sets $f_! : X_s \to X_{s'}$ given by $f_! = \ell|_{\{1\}\times X_s}$.

That is, the data of a left fibration $p : X \to S$ seems to determine a functor $S \to \mathcal{S}$ to the ∞-category of spaces, sending each vertex s to the Kan complex X_s. Indeed, an initial result is the following.

Proposition 4.2.2 ([Lur09, 2.1.1.4])**.** *Let $p : X \to S$ be a left fibration of simplicial sets. The construction described above determines a functor between homotopy categories, $\mathsf{h}S \to \mathsf{h}\mathcal{S}$.*

We would like to upgrade this result to an ∞-categorical statement. Roughly, we want our ∞-categorical Grothendieck construction to establish a correspondence

$$\left\{ \begin{array}{c} \text{Left fibrations of} \\ \infty\text{-categories} \\ p : \mathcal{C} \to \mathcal{D} \end{array} \right\} \overset{\approx}{\longleftrightarrow} \left\{ \begin{array}{c} \text{Functors} \\ \chi : \mathcal{D} \to \mathcal{S} \end{array} \right\},$$

and dually

$$\left\{ \begin{array}{c} \text{Right fibrations of} \\ \infty\text{-categories} \\ p : \mathcal{C} \to \mathcal{D} \end{array} \right\} \overset{\approx}{\longleftrightarrow} \left\{ \begin{array}{c} \text{Functors} \\ \chi : \mathcal{D}^{\mathsf{op}} \to \mathcal{S} \end{array} \right\} \tag{4.2.3}$$

But how do we make sense of this for left fibrations of simplicial sets which are not ∞-categories? And how do we obtain a left fibration from a functor $\chi : \mathcal{D} \to \mathcal{S}$? This leads us to the straightening and unstraightening functors.

4.3. **Straightening and unstraightening (unmarked)**

In this section, we take a first look at the straightening and unstraightening constructions. In the next section, we will discuss their more elaborate "marked" variants. These marked versions will be necessary to make the strongest possible ∞-categorical generalization of the Grothendieck construction. However, we are still able to obtain some nice results in the unmarked case which will serve as intuition in the next section.

Our goal in this section is the following.

Theorem 4.3.1 (Straightening-unstraightening (unmarked) [Lur09, 2.2.1.2])**.** *Let $S \in \mathsf{Set}_\Delta$ be a simplicial set, $\mathcal{C} \in \mathsf{Cat}_\Delta$ a simplicial category, and $\varphi : \mathfrak{C}[S] \to \mathcal{C}^{\mathsf{op}}$ a functor of simplicial categories. There is a Quillen adjunction*

$$(\mathsf{Set}_\Delta)_{/S} \underset{\mathsf{Un}_\varphi}{\overset{\mathsf{St}_\varphi}{\rightleftarrows}} \mathsf{Set}_\Delta^{\mathcal{C}} \tag{4.3.2}$$

where $(\mathsf{Set}_\Delta)_{/S}$ has the contravariant model structure (Remark 3.4.4) and $\mathsf{Set}_\Delta^{\mathcal{C}}$ the projective model structure (Theorem 3.2.3) with respect to the Kan model structure on Set_Δ (Theorem 3.3.1). If φ is a categorical equivalence (Theorem 3.3.3), then (4.3.2) is a Quillen equivalence.

We now construct the straightening functor, St_φ.

Construction 4.3.3 (Straightening functor (unmarked) [Lur09, 2.2.1])**.** Let $S \in \mathsf{Set}_\Delta$ be a simplicial set, $\mathcal{C} \in \mathsf{Cat}_\Delta$ a simplicial category, and $\varphi : \mathfrak{C}[S] \to \mathcal{C}^{\mathsf{op}}$ a functor of simplicial categories.

Let $p : X \to S$ be a map of simplicial sets (an object of $(\mathsf{Set}_\Delta)_{/S}$), let $i : X \hookrightarrow X^{\triangleright}$ be the inclusion, and let $\psi : \mathfrak{C}[X] \to \mathcal{C}^{\mathsf{op}}$ be the composition $\varphi \circ \mathfrak{C}[p]$. Denote the cone point of $X^{\triangleright}$ by $*$. We can form the pushout of simplicial categories

$$\begin{tikzcd} \mathfrak{C}[X] \arrow[r, "\psi"] \arrow[d, "\mathfrak{C}[i]"'] \arrow[dr, phantom, "\ulcorner"] & \mathcal{C}^{\mathsf{op}} \arrow[d] \\ \mathfrak{C}[X^{\triangleright}] \arrow[r] & \mathcal{C}^{\mathsf{op}}_X . \end{tikzcd} \tag{4.3.4}$$

The simplicial category $\mathcal{C}^{\mathsf{op}}_X$ is naturally viewed as a correspondence (in the sense of Definition 2.2.4) from $\mathcal{C}^{\mathsf{op}}$ to $\{v\}$ when equipped with the functor $F : \mathcal{C}^{\mathsf{op}}_X \to \Delta^1$ sending every object of $\mathcal{C}^{\mathsf{op}}$ to 0 and $*$ to 1. Note that, given an object c of $\mathcal{C}^{\mathsf{op}}$, we can identify the simplicial set $\mathsf{Maps}_{\mathcal{C}^{\mathsf{op}}_X}(c, *)$ as the (nerve of the) fiber of ψ over c:

$$\begin{tikzcd} \mathsf{Maps}_{\mathcal{C}^{\mathsf{op}}_X}(c, *) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathfrak{C}[X] \arrow[d, "\psi"] \\ \{c\} \arrow[r] & \mathcal{C}^{\mathsf{op}} . \end{tikzcd}$$

That is, morphisms from c to $*$ in $\mathcal{C}^{\mathsf{op}}_X$ exactly correspond to the vertices of $\mathfrak{C}[X]$ which are sent to c by ψ. We use this to define the *straightening functor* to be the functor sending p to the functor represented by $*$:

$$\begin{aligned} \mathsf{St}_\varphi(p) : \mathcal{C} &\longrightarrow \mathsf{Set}_\Delta \\ c &\longmapsto \mathsf{Maps}_{\mathcal{C}^{\mathsf{op}}_X}(c, *). \end{aligned}$$

As in Construction 4.1.4, establishing the functoriality of St_φ is slightly more subtle than this. To fix this, we instead obtain mapping objects in $\mathcal{C}^{\mathsf{op}}_X$ as the pullbacks

$$\begin{tikzcd} \mathsf{Maps}_{\mathcal{C}^{\mathsf{op}}_X}(d, *) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathfrak{C}[X] \arrow[d, "\psi"] \\ (\mathcal{C}^{\mathsf{op}})_{d/} \arrow[r, "U"] & \mathcal{C}^{\mathsf{op}} \end{tikzcd}$$

so that given a morphism $f : c \to d$ in $\mathcal{C}$, we obtain a natural map of simplicial sets $\mathsf{Maps}_{\mathcal{C}^{\mathsf{op}}_X}(c, *) \to \mathsf{Maps}_{\mathcal{C}^{\mathsf{op}}_X}(d, *)$ as in the following diagram:

$$\begin{tikzcd} \mathsf{Maps}_{\mathcal{C}^{\mathsf{op}}_X}(d, *) \arrow[dr, dashed] \arrow[drr, bend left] \arrow[ddr] & & \\ & \mathsf{Maps}_{\mathcal{C}^{\mathsf{op}}_X}(d, *) \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathfrak{C}[X] \arrow[d, "\psi"] \\ (\mathcal{C}^{\mathsf{op}})_{c/} \arrow[r, "(f^{\mathsf{op}})^*"] & (\mathcal{C}^{\mathsf{op}})_{d/} \arrow[r, "U"] & \mathcal{C}^{\mathsf{op}} . \end{tikzcd}$$

Similarly, morphisms of simplicial sets over S also induce limit comparison maps. Thus, we view St_φ as a functor from $(\mathsf{Set}_\Delta)_{/S}$ to $\mathsf{Set}^{\mathcal{C}}_\Delta$.

To be concrete, let us compute what St_φ does on vertices and edges of a simplicial set X over S.

Example 4.3.5 (Straightening functor on vertices [Lur09, 3.2.1])**.** Let x be a vertex of X, and let $c = \psi(x)$ be the corresponding object of $\mathcal{C}$. By definition, x is a map $x : \Delta^0 \to X$. Thus, we obtain an edge $x \star \mathsf{id}_{\Delta^0}$

of $X^{\triangleright}$ via the following diagram.

$$\begin{tikzcd} \Delta^0 \star \Delta^0 \arrow[r, "x \star \mathsf{id}_{\Delta^0}"] \arrow[d, "\cong"] & X \star \Delta^0 = X^{\triangleright} \\ \Delta^1 \arrow[ur] & \end{tikzcd}$$

Applying the map $\mathfrak{C}[X^{\triangleright}] \to \mathcal{C}_X^{\mathsf{op}}$ of (4.3.4) to this edge, we obtain a morphism $c \to *$ in $\mathcal{C}_X^{\mathsf{op}}$ which we identify with a vertex $\tilde{c}$ of

$$\mathsf{St}_\varphi(X)(c) = \mathsf{Maps}_{\mathcal{C}_X^{\mathsf{op}}}(c, *).$$

Example 4.3.6 (Straightening functor on edges [Lur09, 3.2.1])**.** Let $f : x \to y$ be an edge of X, and let $c = \psi(x)$, $d = \psi(y)$ and $F = \psi(f) : c \to d$ be the corresponding vertices and edge of $\mathcal{C}^{\mathsf{op}}$. By definition, f is a map $f : \Delta^1 \to X$. We obtain a 2-simplex in X as in the following diagram.

$$\begin{tikzcd} \Delta^1 \star \Delta^0 \arrow[r, "f \star \mathsf{id}_{\Delta^0}"] \arrow[d, "\cong"'] & X \star \Delta^0 = X^{\triangleright} \\ \Delta^2 \arrow[ur] & \end{tikzcd}$$

Applying $\mathfrak{C}[-]$ and composing with the map $\mathfrak{C}[X^{\triangleright}] \to \mathcal{C}_X^{\mathsf{op}}$ of (4.3.4), we obtain $\mathfrak{C}[\Delta^2] \to \mathcal{C}_X^{\mathsf{op}}$, corresponding to a diagram

$$\begin{tikzcd} c \arrow[rr, "F"] \arrow[dr, "\tilde{x}"'] & & d \arrow[dl, "\tilde{y}"] \\ & * & \end{tikzcd}$$

in $\mathcal{C}_X^{\mathsf{op}}$ along with an edge

$$\tilde{f} : \tilde{x} \to \tilde{d} \circ F = F^* \tilde{d}$$

in

$$\mathsf{St}_\varphi(X)(c) = \mathsf{Maps}_{\mathcal{C}_X^{\mathsf{op}}}(c, *).$$

One can check that St_φ as defined above preserves colimits, and so admits a left adjoint Un_φ which we call the *unstraightening functor*. It turns out that we can describe Un_φ pretty well in nice cases.

Example 4.3.7 ([Lur09, 2.2.1])**.** Let S be a simplicial set and let $\varphi : \mathfrak{C}[S] \to \mathcal{C}^{\mathsf{op}}$ be a simplicial functor, and form the associated unstraightening functor

$$\mathsf{Un}_\varphi : \mathsf{Set}_\Delta^{\mathcal{C}} \to (\mathsf{Set}_\Delta)_{/S}.$$

Let $\mathcal{F} : \mathcal{C} \to \mathsf{Set}_\Delta$ be a functor taking values in Kan complexes. Then $\mathsf{Un}_\varphi \mathcal{F}$ is a right fibration $X \to S$ of simplicial sets with fiber over a vertex s given by the Kan complex $\mathcal{F}(\varphi(s))$.

Finally, in the case where $p : S \to S$ and $\varphi : \mathfrak{C}[S] \to \mathfrak{C}[S] = \mathcal{C}^{\mathsf{op}}$ are both identity morphisms, we arrive at a partial resolution of the goal laid out in (4.2.3).

Proposition 4.3.8 ([Lur09, 2.2.3.11])**.** *For every simplicial set S, the unstraightening functor Un_S associated to the identity functor $\mathfrak{C}[S] \to \mathfrak{C}[S]$ induces an equivalence*

$$(\mathsf{Set}_\Delta^{\mathfrak{C}[S]^{\mathsf{op}}})^\circ \to \mathsf{RFib}(S) \subset (\mathsf{Set}_\Delta)_{/S}$$

where on the left we have functors $\mathfrak{C}[S]^{\mathsf{op}} \to \mathsf{Set}_\Delta$ which are bifibrant with respect to the projective model structure, and on the right we have the full subcategory of right fibrations to S.

In particular, if we take S to be an ∞-category $\mathcal{D}$, then any functor $\mathcal{D}^{\mathsf{op}} \to \mathcal{S}$ is bifibrant since Kan complexes are bifibrant in the Kan model structure on Set_Δ, and so unstraightens to a right fibration $\mathcal{C} \to \mathcal{D}$.

4.4. Cat_∞ and the need for marked simplicial sets

Recall from Theorem 3.3.8 that we can *define* ∞-categories to be precisely the fibrant objects of Set_Δ with respect to the Joyal model structure. Accordingly, we typically want to regard ∞-categories as "the same" if there is a categorical equivalence between them (i.e., a weak equivalence in the Joyal model structure), rather than requiring them to be isomorphic as simplicial sets. The former is a much coarser notion; recall from Proposition 3.3.6 that a map of simplicial sets $w : \mathcal{C} \to \mathcal{D}$ is a categorical equivalence if it is essentially surjective up to homotopy and induces weak homotopy equivalences on mapping spaces:

$$w : \mathsf{Maps}_{\mathcal{C}}(c, c') \xrightarrow{\simeq} \mathsf{Maps}_{\mathcal{D}}(w(c), w(c')).$$

Thus, considering the full subcategory of Set_Δ spanned by the ∞-categories is insufficient for us; we would like to obtain an ∞-category of ∞-categories in which mapping objects "keep track" of the data of categorical equivalences up to all higher homotopy coherence. The answer to this is the following.

Definition 4.4.1 ([Lur09, 3.0.0.1])**.** Define $\mathsf{Cat}_\infty^\Delta$ to be the simplicial category defined as follows:

(i) the objects of $\mathsf{Cat}_\infty^\Delta$ are (small) ∞-categories, and
(ii) given ∞-categories $\mathcal{C}$ and $\mathcal{D}$, we define the simplicial set

$$\mathsf{Maps}_{\mathsf{Cat}_\infty^\Delta}(\mathcal{C}, \mathcal{D}) = \operatorname{colim}_{K \subset \mathsf{Fun}(\mathcal{C},\mathcal{D})} K$$

in which the colimit ranges over all Kan complexes K contained in the ∞-category $\mathsf{Fun}(\mathcal{C}, \mathcal{D})$. That is, $\mathsf{Maps}_{\mathsf{Cat}_\infty^\Delta}(\mathcal{C}, \mathcal{D})$ is the "largest" Kan complex contained in $\mathsf{Fun}(\mathcal{C}, \mathcal{D})$. (This is sometimes called the *core* of $\mathsf{Fun}(\mathcal{C}, \mathcal{D})$.)

Definition 4.4.2 ([Lur09, 3.0.0.1])**.** We define the *∞-category of (small) ∞-categories*, Cat_∞, to be the simplicial nerve of $\mathsf{Cat}_\infty^\Delta$:

$$\mathsf{Cat}_\infty = N(\mathsf{Cat}_\infty^\Delta).$$

Remark 4.4.3 ([Lur09, 3.0.0.2])**.** A major advantage of working with $\mathsf{Cat}_\infty^\Delta$ is that composition is strictly associative and functors can be defined "precisely" at the level of simplicial sets.

In Section 4.3, we came close to resolving the goal of (4.2.3). In what follows, we would like to push this result further to handle functors $\mathcal{D}^{\mathsf{op}} \to \mathsf{Cat}_\infty$. This is a much more subtle problem.

For example, we would like to use the following theorem in order to deal with limits and colimits in Cat_∞ as homotopy limits and homotopy colimits in some appropriate model category.

Theorem 4.4.4 ([Lur09, 4.2.4.8])**.** *Let $\mathcal{A}$ be a combinatorial simplicial model category (Definition 3.2.1, Definition 3.1.6). Then the ∞-category $N(\mathcal{A}^\circ)$ admits all (small) limits and colimits.*

From how we have defined Cat_∞ (Definition 4.4.2), the obvious choice for $\mathcal{A}$ seems to be Set_Δ with the Joyal model structure. However a major complication arises: when Set_Δ is given the usual structure of a simplicial category, the Joyal model structure cannot be made a *simplicial* model structure in the sense of Definition 3.1.6. This is the utility of the Cartesian model structure (Theorem 3.4.6). When Set_Δ^+ (or rather, $(\mathsf{Set}_\Delta^+)_{/*}$) is equipped with the Cartesian model structure, there is an equivalence of *simplicial* categories

$$\mathsf{Cat}_\infty^\Delta \simeq (\mathsf{Set}_\Delta^+)^\circ,$$

leading us to alternately define

$$\mathsf{Cat}_\infty = N((\mathsf{Set}_\Delta^+)^\circ).$$

With this definition in place, we are motivated to define *marked* versions of the straightening and unstraightening functors.

4.5. Straightening and unstraightening (marked)

In this section, we will discuss the following marked variant of Theorem 4.3.1.

Theorem 4.5.1 (Straightening-unstraightening (marked) [Lur09, 3.2.0.1])**.** *Let S be a simplicial set, $\mathcal{C}$ a simplicial category, and $\varphi : \mathfrak{C}[S] \to \mathcal{C}^{\mathsf{op}}$ a simplicial functor. There is a Quillen adjunction*

$$(\mathsf{Set}_\Delta^+)_{/S} \underset{\mathsf{Un}_\varphi^+}{\overset{\mathsf{St}_\varphi^+}{\rightleftarrows}} (\mathsf{Set}_\Delta^+)^{\mathcal{C}}$$

where $(\mathsf{Set}_\Delta^+)_{/S}$ is given the Cartesian model structure (Theorem 3.4.6) and $(\mathsf{Set}_\Delta^+)^{\mathcal{C}}$ the projective model structure (Theorem 3.2.3) induced by the Cartesian model structure on Set_Δ^+.

This adjunction is a Quillen equivalence whenever φ is an equivalence of simplicial categories.

We begin by constructing the functor St_φ^+.

Construction 4.5.2 (Straightening functor (marked) [Lur09, 3.2.1])**.** Let S, $\mathcal{C}$, and φ be as in Theorem 4.5.1. We wish to construct the functor

$$\mathsf{St}_\varphi^+ : (\mathsf{Set}_\Delta^+)_{/S} \to (\mathsf{Set}_\Delta^+)^{\mathcal{C}}.$$

That is, for each $(X, \mathcal{E})$ in $(\mathsf{Set}_\Delta^+)_{/S}$, we want to produce a functor

$$\mathsf{St}_\varphi^+(X, \mathcal{E}) : \mathcal{C} \to \mathsf{Set}_\Delta^+ .$$

For a given object c of $\mathcal{C}$, we define the underlying simplicial set of $\mathsf{St}_\varphi^+(X, \mathcal{E})(c)$ to be $\mathsf{St}_\varphi(X)(c)$, where St_φ is as in Construction 4.3.3.

This leaves us with the problem of producing a set of marked edges in $\mathsf{St}_\varphi(X)(c)$ in a way which is natural in c and $\mathcal{E}$. For each c in $\mathcal{C}$, we define the set $\mathcal{E}_\varphi(c)$ to be the set of all edges in $\mathsf{St}_\varphi(X)(c)$ of the form $G^*\tilde{f}$ where $f : y \to z$ is an edge of X, $\tilde{f}$ is the edge of $\mathsf{St}_\varphi(\psi(y)) = \mathsf{Maps}_{\mathcal{C}_X^{\mathsf{op}}}(\psi(y), *)$ obtained as in Example 4.3.6, and G is an edge of $\mathsf{Maps}_{\mathcal{C}^{\mathsf{op}}}(c, \psi(y))$. Concretely, if $G : \alpha \to \beta$, then $G^*\tilde{f}$ is an edge $\tilde{y} \circ \beta \to (\psi(f)^*\tilde{z}) \circ \alpha$, obtained as in the following diagram.

[This diagram won't compile for some reason...]

The sets $\{\mathcal{E}_\varphi(c)\}$ can be characterized as the smallest collection of of sets containing $\tilde{f}$ for every f in $\mathcal{E}$ which depend functorially on c.

Thus, we define

$$\mathsf{St}_\varphi^+(X, \mathcal{E})(c) = (\mathsf{St}_\varphi(X)(c), \mathcal{E}_\varphi(c)).$$

5. Applications to Cat_∞

[Sorry, I got busy with other things and ran out of time to write this section. Hopefully I'll come back to it someday. Of course, it is all based on [Lur09, 3.3]]

5.1. Classifying functors to Cat_∞

[Say something about universal (co)Cartesian fibration...]

5.2. Explicit construction of colimits (and limits) in Cat_∞

[The goal is to flesh out this very nice theorem]

Theorem 5.2.1. *Let $p : K^{\mathsf{op}} \to \mathsf{Cat}_\infty$ be a diagram of ∞-categories classifying a Cartesian fibration $X \to K$. Then there is a natural isomorphism*

$$\operatorname{colim} p \simeq X^\natural$$

in hCat_∞.

Remark 5.2.2. In the case where $\mathcal{C}$ is an ∞-category and $F : \mathcal{C}^{\mathsf{op}} \to \mathsf{Cat}_\infty$ is a functor, then the corresponding Cartesian fibration $\mathcal{E} \to \mathcal{C}$ is also a functor of ∞-categories, and the above theorem becomes the following:

$$\operatorname{colim} F \simeq \mathcal{E}[\mathrm{Cart}^{-1}]$$

where on the right hand side we have the localization of $\mathcal{E}$ at the class of Cartesian edges.

Dually, we also have that

$$\lim F \simeq \Gamma_{\mathrm{Cart}}(\mathcal{E})$$

where the right hand side is the ∞-category of Cartesian sections of $\mathcal{E} \to \mathcal{C}$.

[Make it clear how that remark is true and explain what Cartesian sections are.]

6. Colimits

[This section also needs to be filled in... the reference is [Lur09, 4.2 and 4.3]]

6.1. ∞-Categorical colimits as homotopy colimits

The following essential theorem which basically says that limits and colimits in ∞-categories can be realized as homotopy limits and colimits in some appropriate simplicial model category.

Theorem 6.1.1 ([Lur09, 4.2.4.1])**.** *Let $\mathcal{J}$ and $\mathcal{C}$ be fibrant simplicial categories, and $F : \mathcal{J} \to \mathcal{C}$ a simplicial functor. Let $c \in \mathcal{C}$ be an object, and suppose we have a compatible family of maps*

$$\{\eta_I : F(I) \to c\},$$

i.e., an extension of F to a functor $F^{\triangleright} : \mathcal{J}^{\triangleright}$ sending the cone point $*$ to c. Then the following are equivalent.

(i) The functor $F^{\triangleright}$ witnesses c as a colimit of the diagram F;
(ii) $N(F^{\triangleright}) : N(\mathcal{J}^{\triangleright}) \to N(\mathcal{C})$ is a colimit diagram in $\mathcal{C}$.

A consequence is the following abstract construction of limits and colimits in ∞-categories. Compare this to the explicit construction discussed in Section 5.2.

Corollary 6.1.2 ([Lur09, 4.2.4.8])**.** *Let $\mathcal{A}$ be a combinatorial simplicial model category. The associated ∞-category $N(\mathcal{A}^{\circ})$ admits all limits and colimits.*

6.2. Relative colimits and left Kan extensions

Index

$S^\flat$, 5
$S^\natural$, 5
$S^\sharp$, 5
$\mathfrak{C}[-]$, 4
∞-category, 2
  ∞-category of, 24
  equivalence of, 17
  parameterized, 8
∞-groupoid, 3

acyclic
  cofibration, 14
  fibration, 14
anodyne, 7
  inner, 7
  left, 7
  right, 7

Cartesian closed, 6
categorical equivalence, 16
classifying functor, 25
cofibered, 18
cofibrant, 14
correspondence, 8
cosimplicial
  object, 2
cotensored, 15

derived functor
  left, 14
  right, 14

edge
  Cartesian, 10
  coCartesian, 11
  locally Cartesian, 11

fibrant, 14
fibration
  Cartesian, 11
  categorical, 16
  coCartesian, 11
  inner, 7
  Kan, 7
  left, 7
  locally Cartesian, 12
  right, 7
  trivial, 6

geometric realization, 4
Grothendieck construction, 18

homotopy category, 14
homotopy colimit, 26
homotopy equivalence, 7
  weak, 7
horn, 2
  filling property, 3
  inner, 2
  outer, 2

Kan complex, 3
Kan extension, 26

lifting property
  left, 6
  right, 6
  unique, 11

mapping space, 12
marked, 5
model category
  combinatorial, 15
  enriched, 14
  left proper, 14
  right proper, 14
  simplicial, 14
model structure
  Cartesian, 18
  coCartesian, 18
  contravariant, 17
  covariant, 17
  injective, 15
  Joyal, 16
  Kan, 16
  on Cat_Δ, 16
  projective, 15

nerve
  classical, 4
  homotopy-coherent, 5

presentable, 15

Quillen adjunction, 14
Quillen equivalence, 14

simplicial
  category, 3
  model category, 14
  nerve, 5
  object, 2
  set, 2
  thickening functor, 4
singular chains, 4
six functor formalism, 9
straightening
  classical, 19
  marked, 25
  unmarked, 22

tensored, 14
topological category, 3

unstraightening
  classical, 19
  unmarked, 23

weakly saturated, 15

References

[Lur09] Jacob Lurie, *Higher topos theory*, Princeton, 2009.
[Lur25] ———, *Kerodon*, https://kerodon.net/, 2025.
[Rhi14] Emily Rhiel, *Categorical homotopy theory*, Cambridge University Press, 2014.
[Sch26] Peter Scholze, *Six-functor formalisms*, arXiv:2510.26269v2, 2026.

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